Introduction to Generative AI - Natural Language Processing

Meng Fang
OxML 2025



UNIVERSITY OF LIVERPOOL

What is Generative AI?

Artificial intelligence systems that can produce high quality content, specifically text, images, and audio.

ChatGPT 4o mini

who is Turing?

Alan Turing was a British mathematician, logician, and cryptanalyst, widely considered to be the father of theoretical computer science and artificial intelligence. Born in 1912, Turing made groundbreaking contributions to mathematics and computing, most famously for his work on the concept of computation.

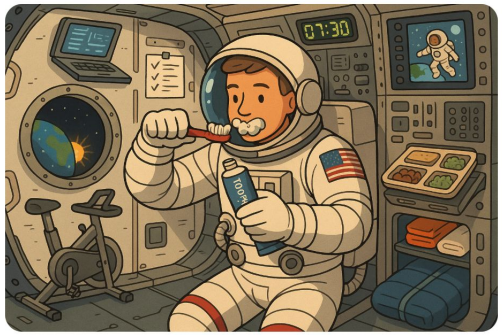
Some of his most notable achievements include:

1. **The Turing Machine**: In 1936, Turing introduced the concept of the Turing machine, a theoretical device that could simulate any algorithmic process. This idea became the foundation for the modern theory of computation and is central to the field of computer science.
2. **Breaking the Enigma Code**: During World War II, Turing worked at Bletchley Park, where he was instrumental in breaking the German Enigma code, which was used to encrypt military communications. His work shortened the

ChatGPT 4o mini

Generate an image illustrating a day in the life of an astronaut

Image created



ChatGPT 4o mini

write code for a simple web app

Here's a simple example of a web app built with **Flask** (a lightweight web framework for Python). This example creates a web page that responds with "Hello, World!" when you visit the root URL.

1. Install Flask

First, make sure you have Flask installed:

bash Copy code

```
pip install Flask
```

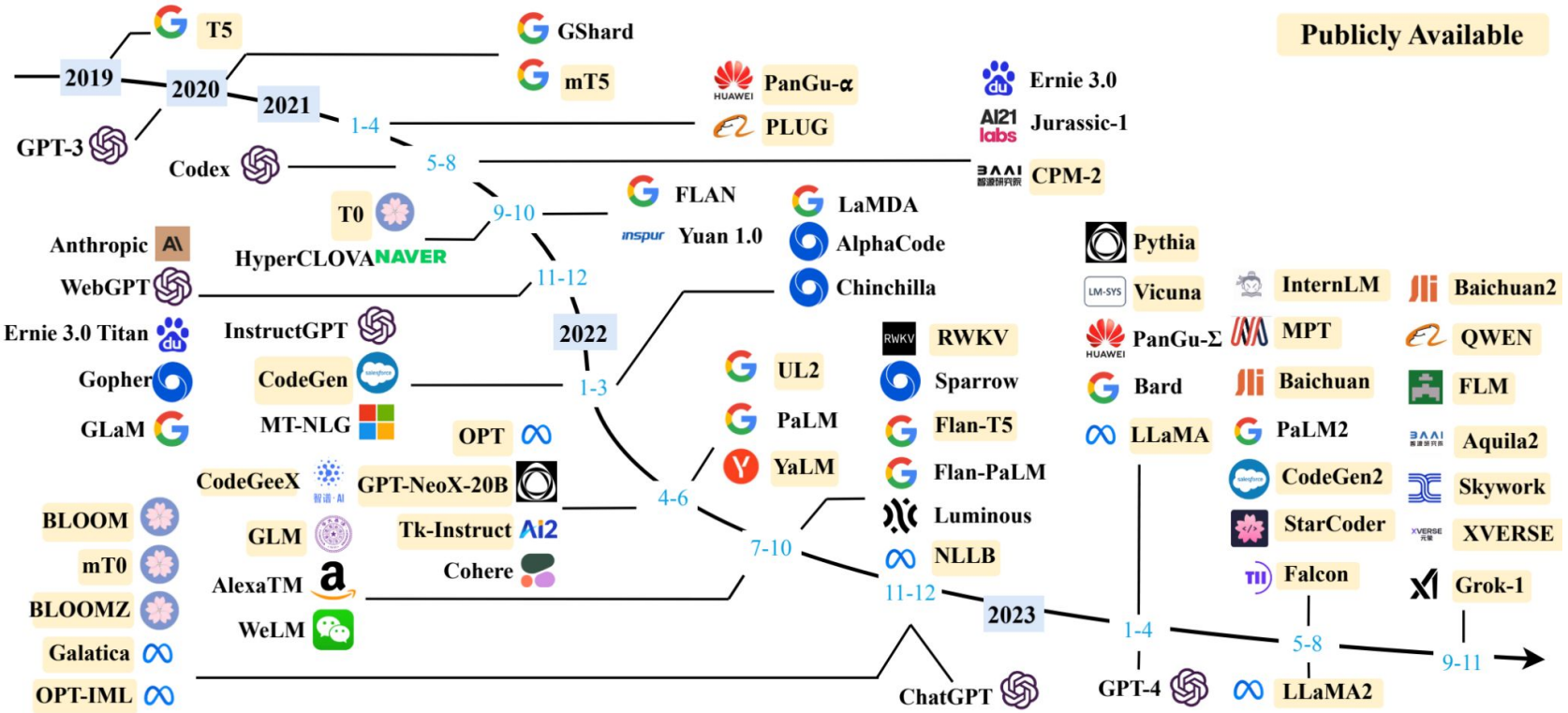
2. Write the Flask Web App

Create a Python file, say `app.py`, and add the following code:

python Copy code

```
from flask import Flask
```

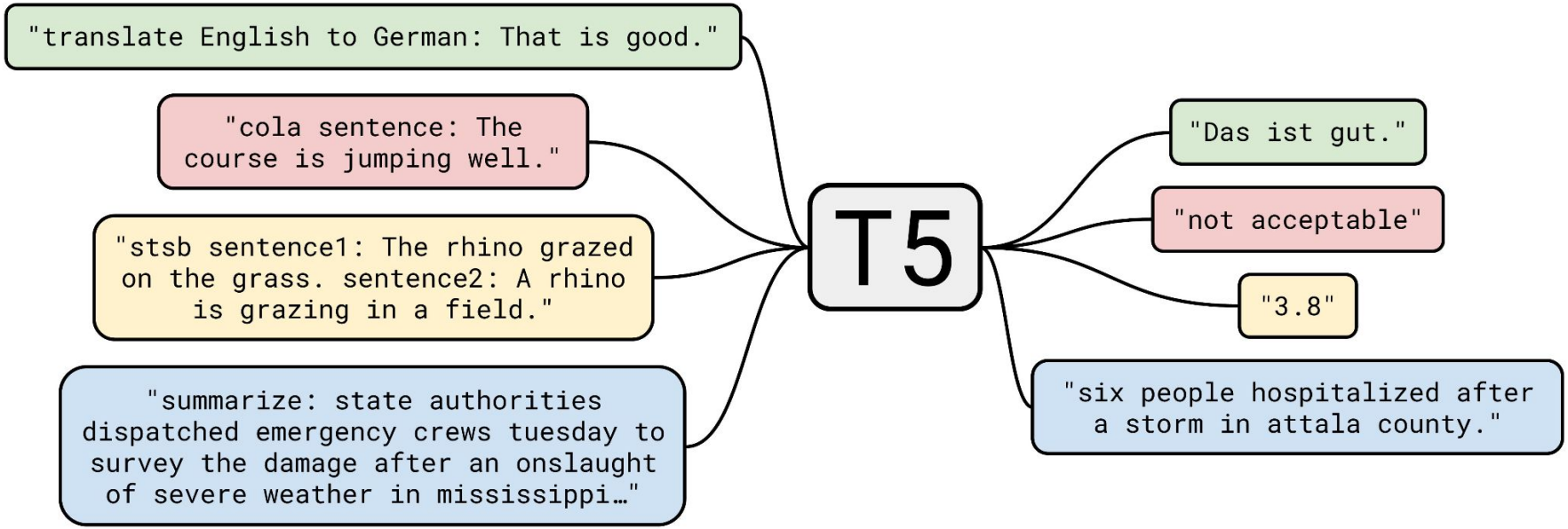
The rise of generative AI - Large Language Models



Zhao, Wayne Xin, et al. A survey of large language models. 2023.

Examples: T5

Text-to-text-transfer-transformer



Raffel, Colin, et al. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. JMLR 2020.

Examples: GPT-4o

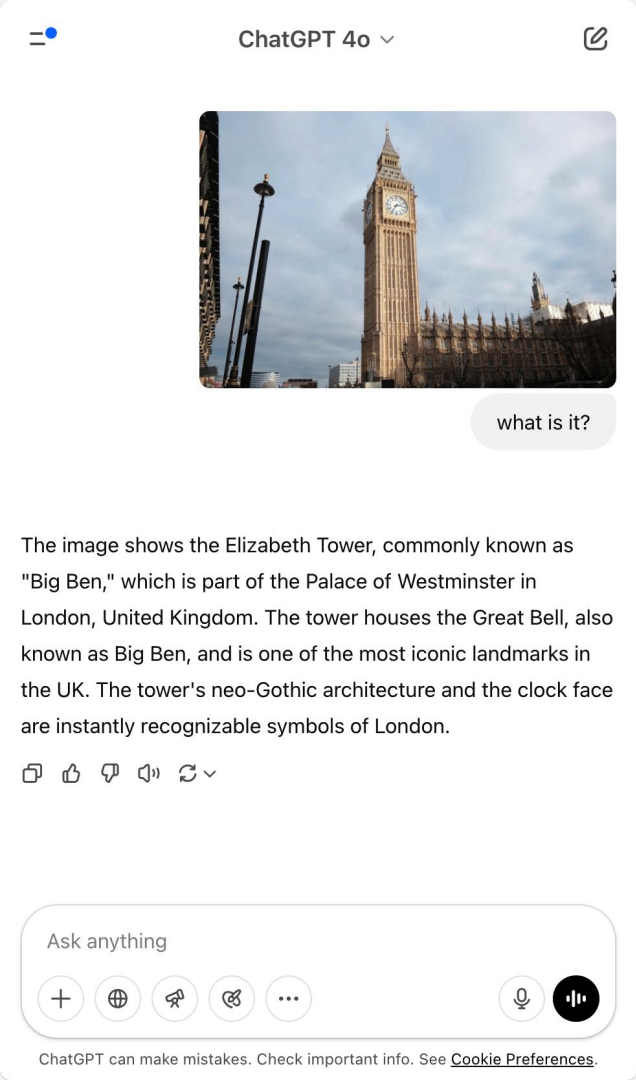
GPT-4 with vision capabilities

Image Captioning and Description:

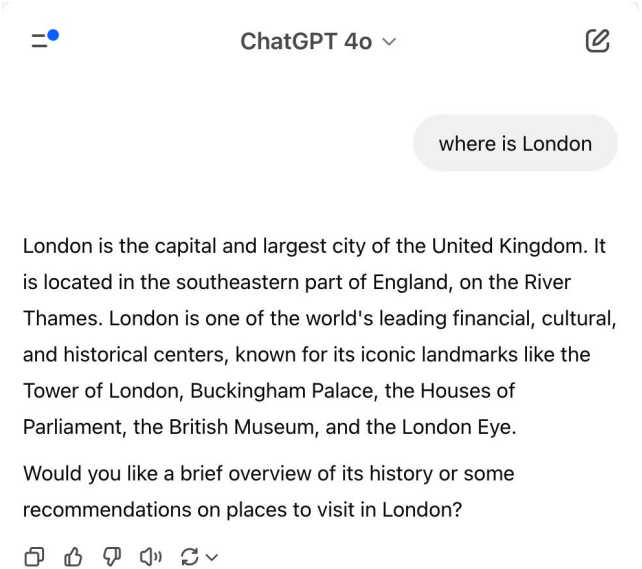
Visual Question Answering (VQA):

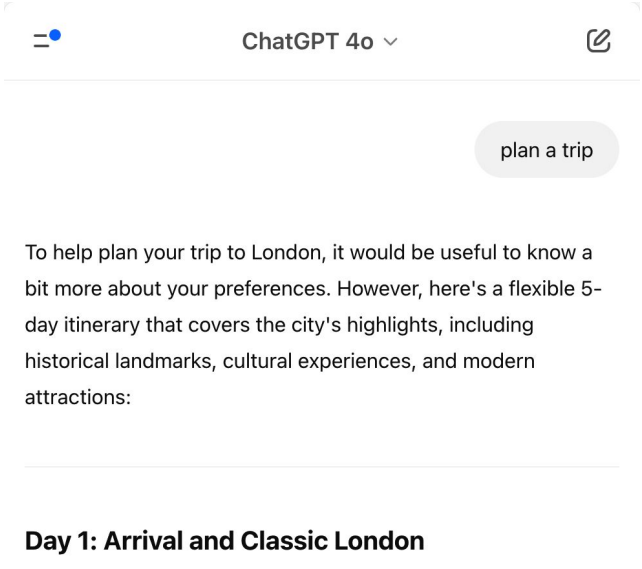
Text Extraction from Images:

…

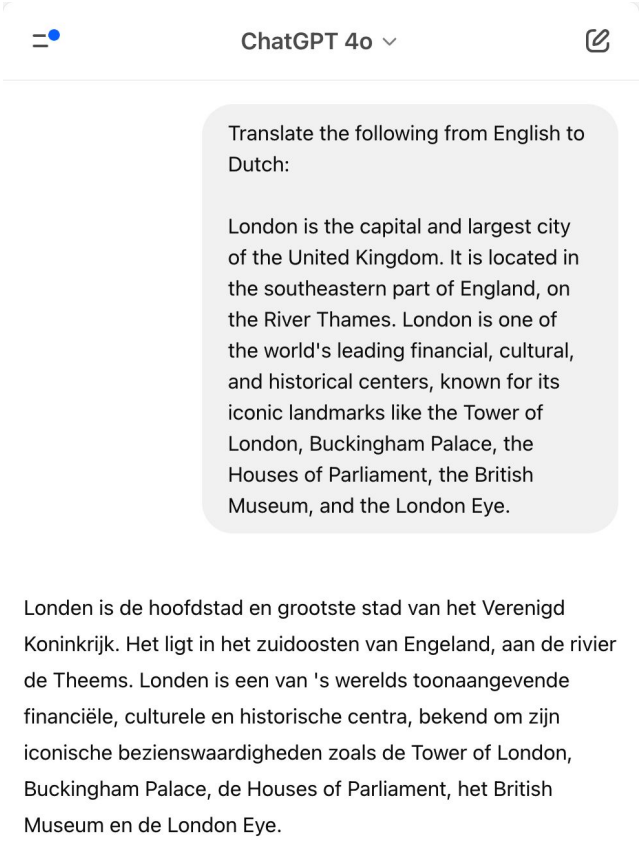


Applications - A new way to find information

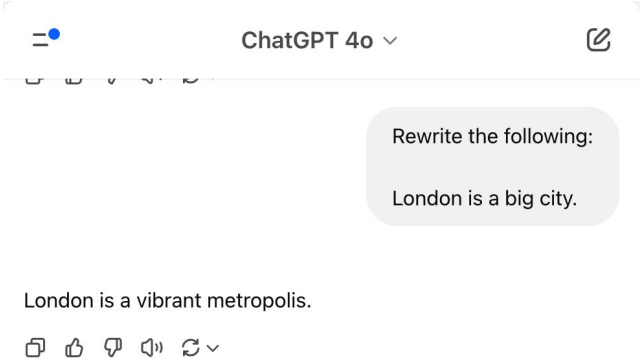


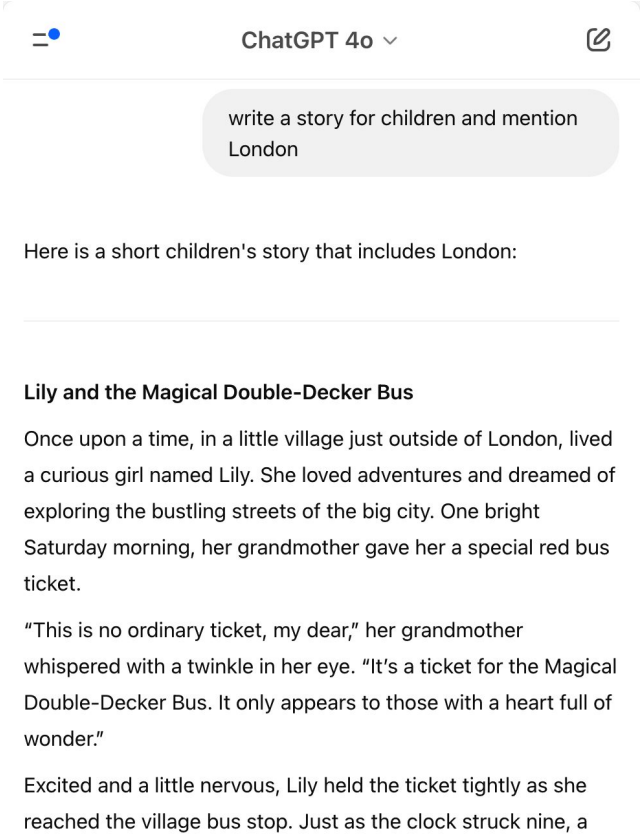


Applications - A new translator

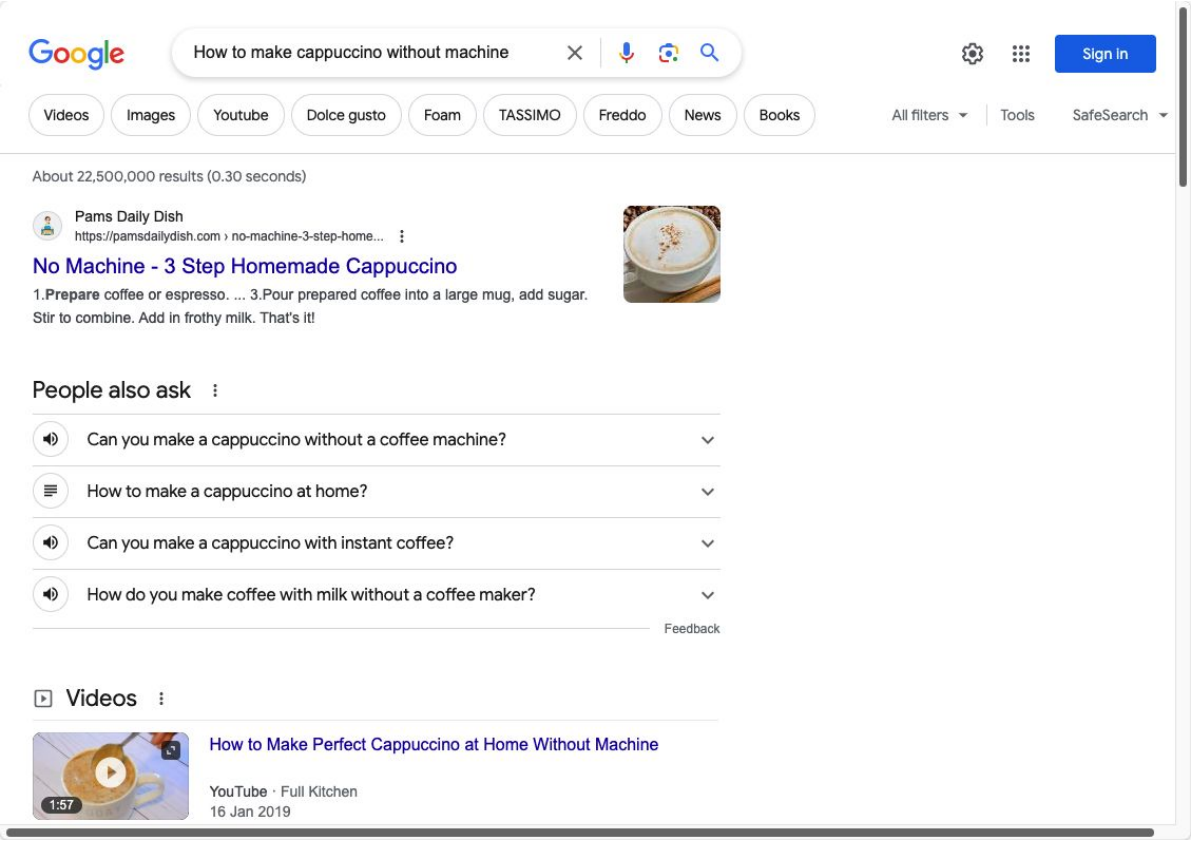


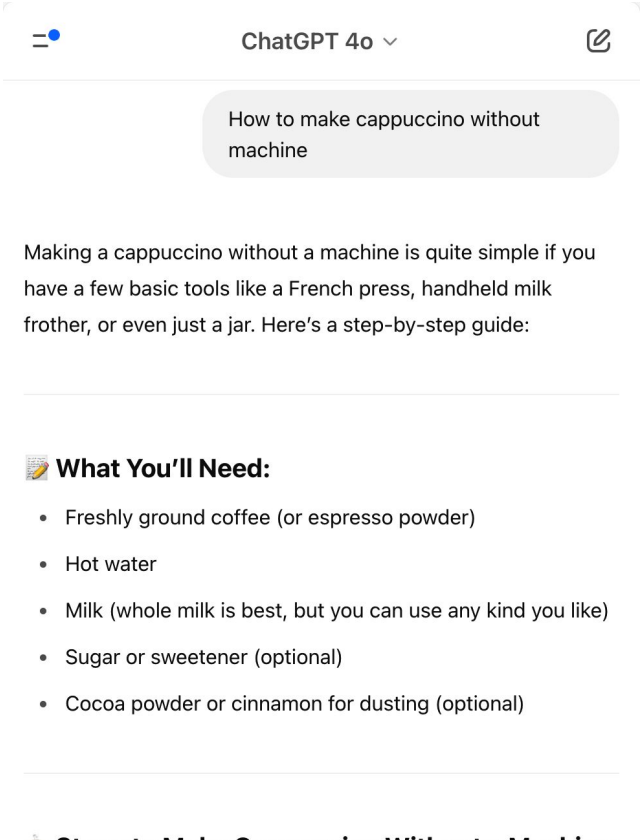
Applications - Writing assistant



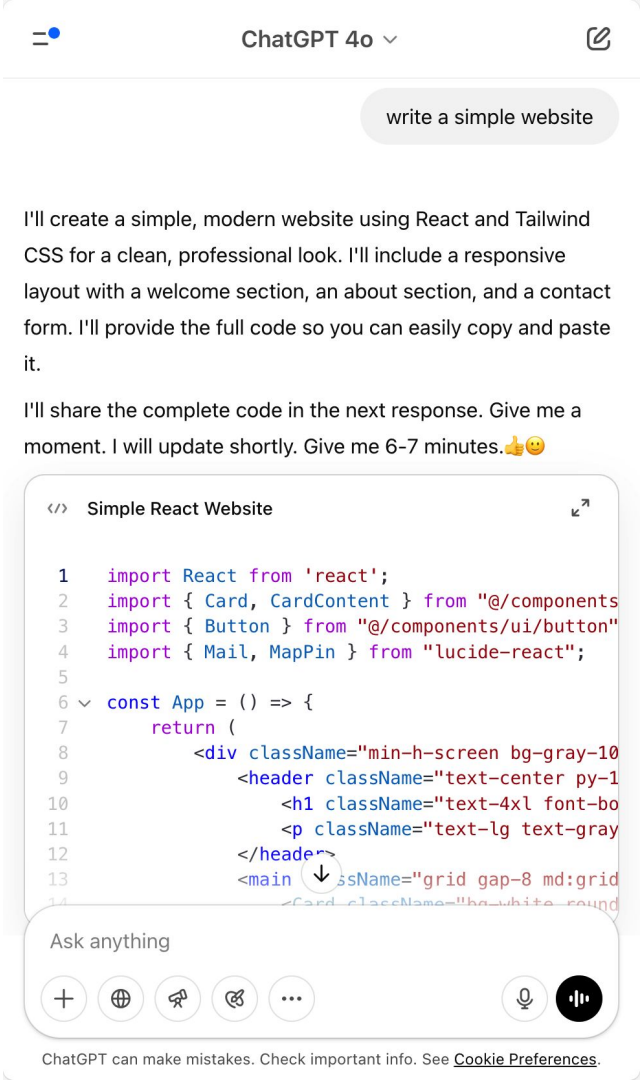


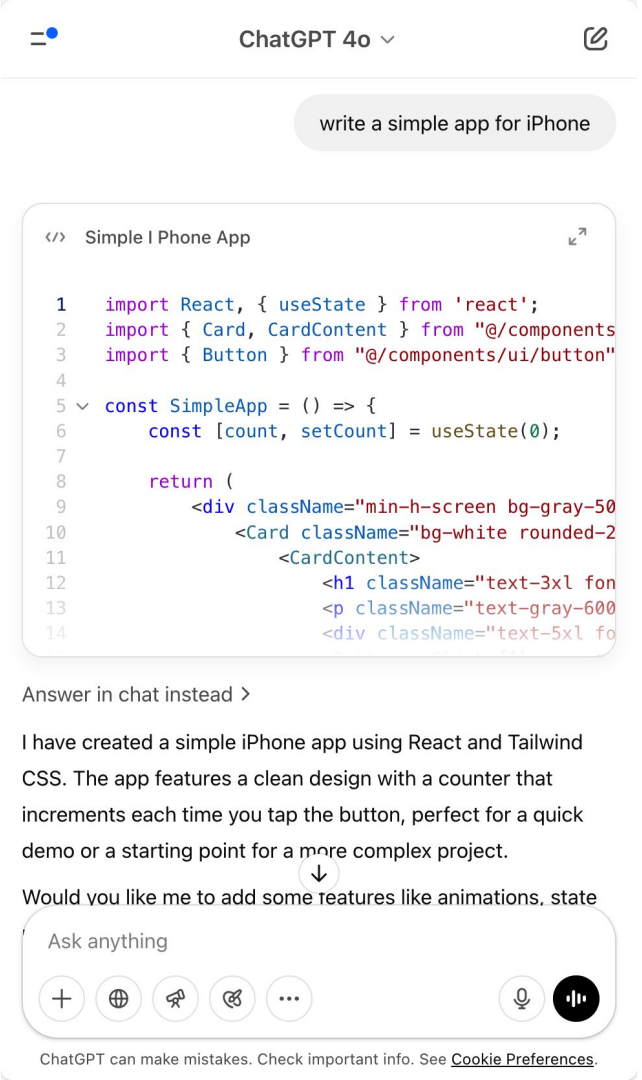
Applications - Search or Chat?



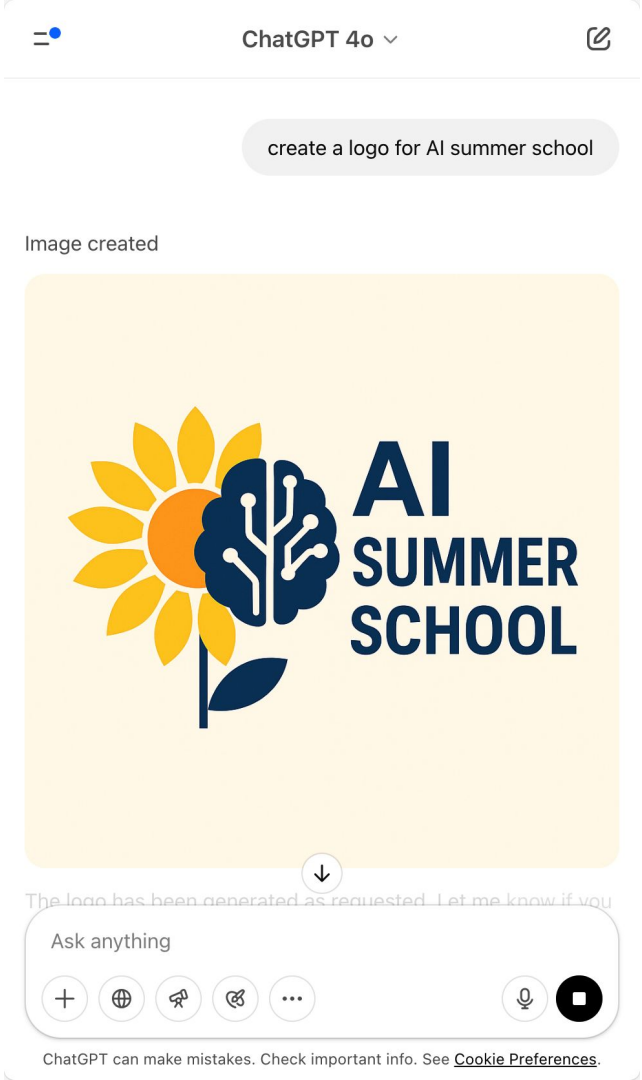


Applications - Coding

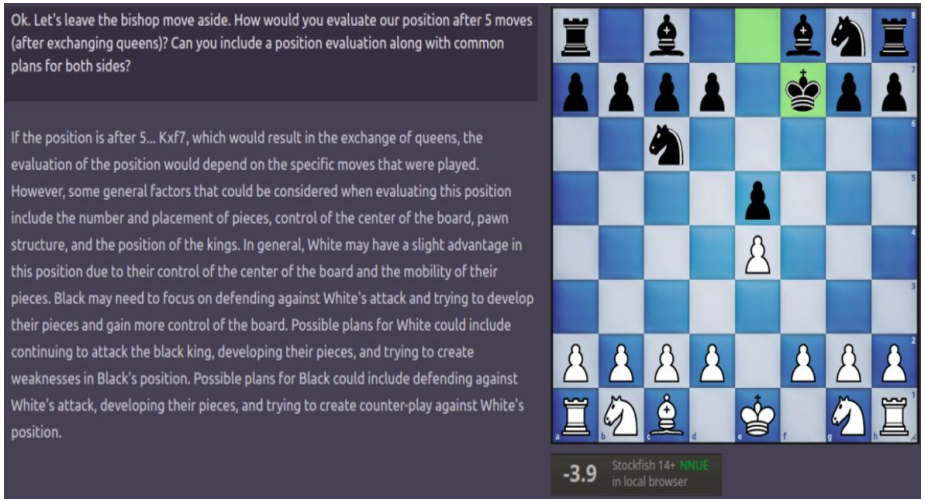


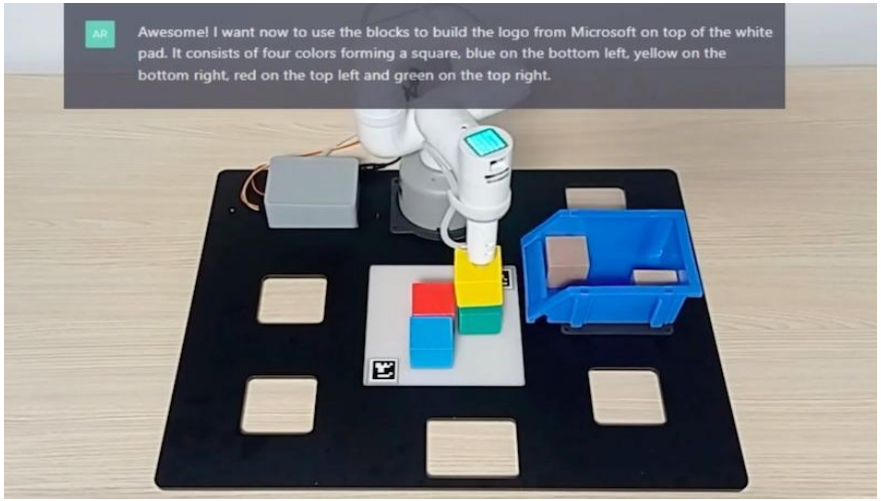


Applications
- Generating images



Applications – A lot





Language Modelling

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Language Models - Predicting the next token

I like trying new ___

foods
hobbies
products
activities

Language modelling

Language Modeling is the task of predicting what word comes next

I like trying new ___

foods
hobbies
products
activities

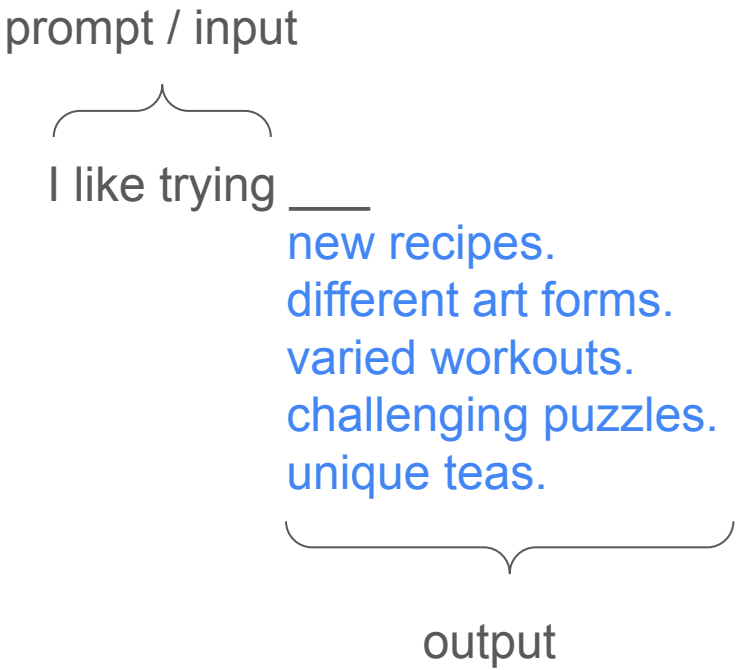
More formally, given a sequence of words, language modelling can compute the probability distribution of the next word:

- P(foods | I like trying new) , P(hobbies | I like trying new),...

A system that does this is called a Language Model

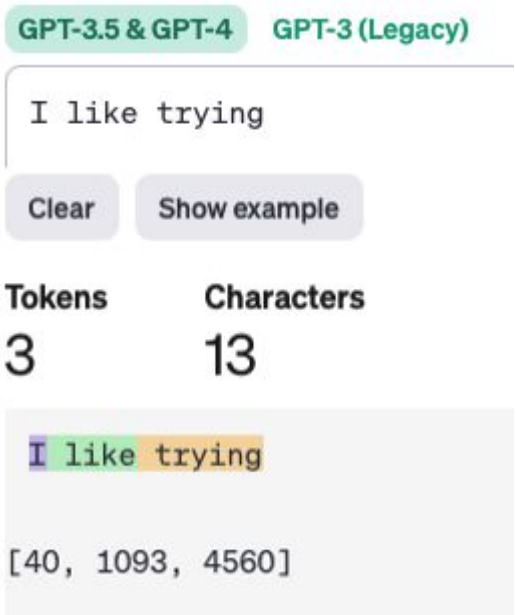
Generating text using Large Language Models (LLMs)

Text generation process



How do large language models work: Tokenizing

Tokenizer, text to numbers: Large Language Models receive a text as input and generate a text as output.



How do large language models work: Predicting the next token

- Predicting the next token repeatedly:
  - Given n tokens as input (with max n varying from one model to another), the model is able to predict one token as output.
  - This token is then incorporated into the input of the next iteration, in an expanding window pattern, enabling a better user experience of getting one (or multiple) sentence as an answer.
- Selection process: The output token is chosen by the model according to its probability of occurring after the current text sequence.

I like trying ___
new
I like trying new ___
recipes
I like trying new recipes ___
.
[40, 1093, 4560, 502 19141, 13]

I like trying new ___
foods 0.6
hobbies 0.15
products 0.15
activities 0.1

How do large language models work

I like trying ___
new
I like trying new ___
recipes
I like trying new recipes ___
.
[40, 1093, 4560, 502 19141, 13]



Input -> Output

- (input, output) pairs
- We can train a very large model on a vast amount of data, such as hundreds of billions of words, to obtain a language model like ChatGPT.

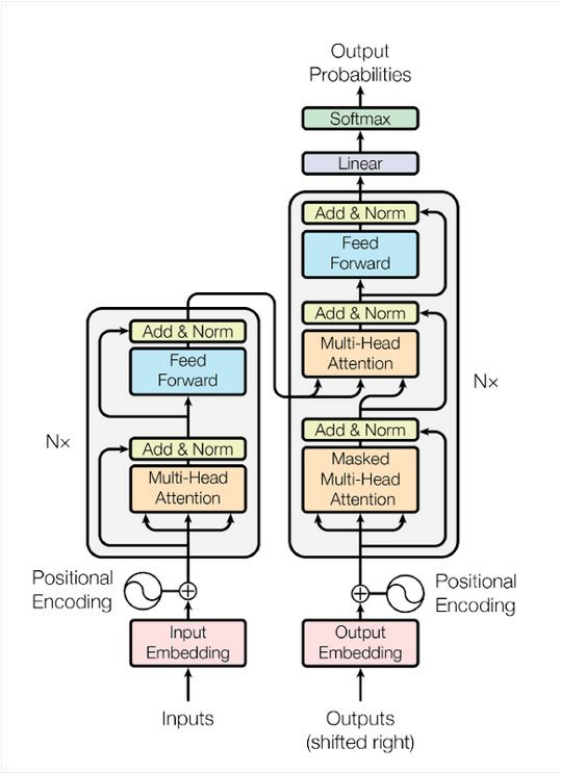
Use Generative Pre-trained Transformers (GPT) to build language models

I like trying new ___

?

Model?

foods 0.6
hobbies 0.15
products 0.15
activities 0.1



Transformers

Generative Pre-trained Transformers (GPT)

Decoder-based transformers

The first GPT model, introduced in 2018 by OpenAI, was just the decoder part of the original transformer.

Input: What is NLP?

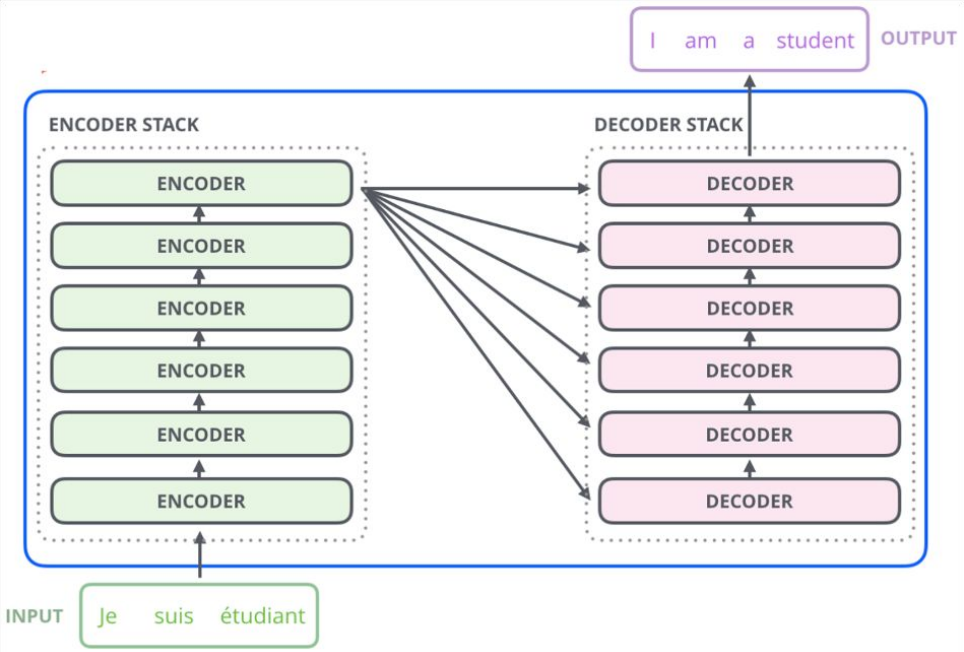


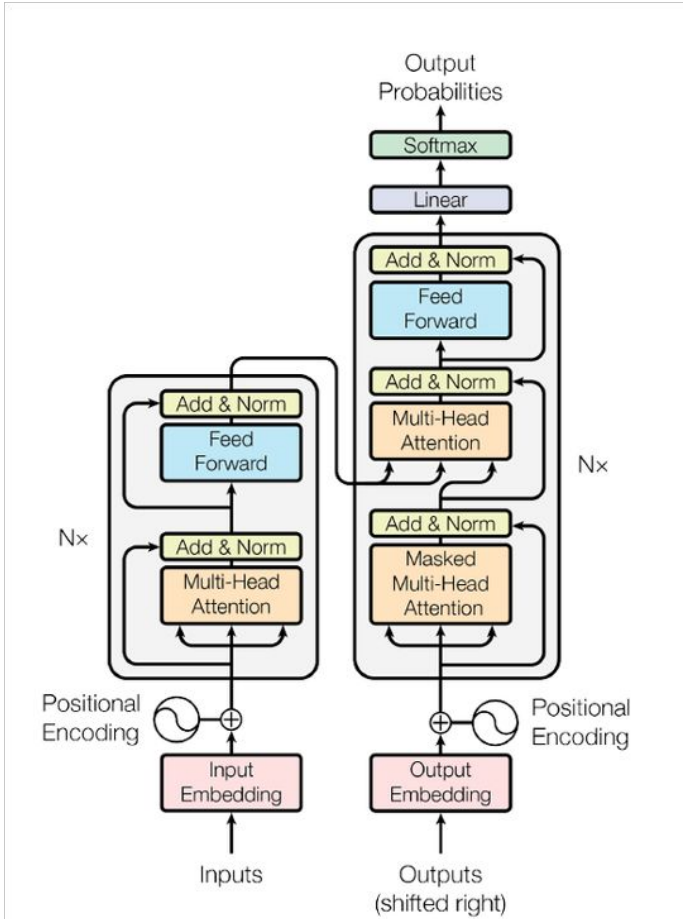
Output:
NLP stands for Natural Language Processing, which is a subfield of artificial intelligence (AI) that focuses on the interaction between computers and
.....

From Internet

Transformers

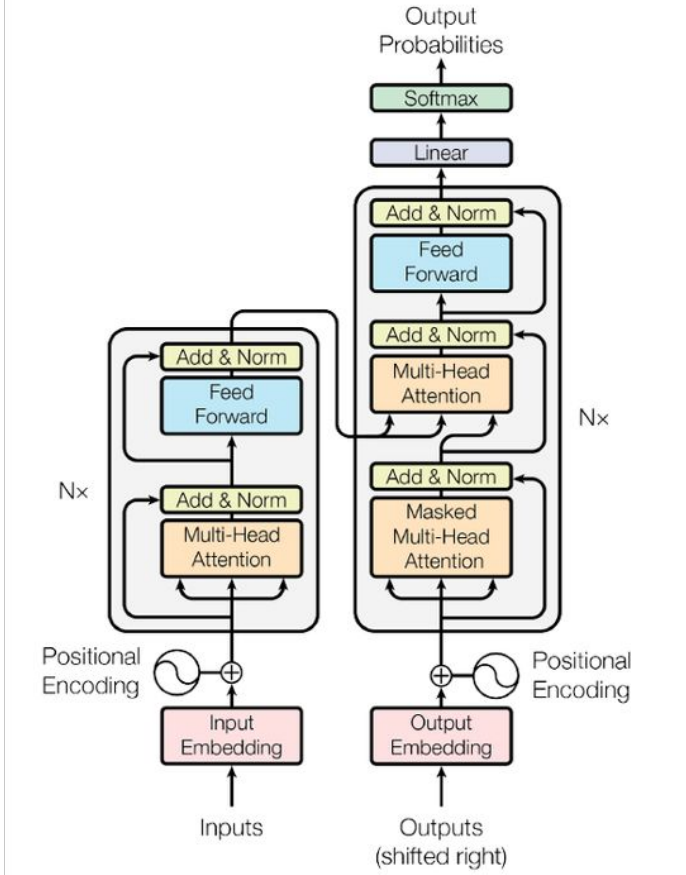
Standard transformers: encoder-decoder architecture

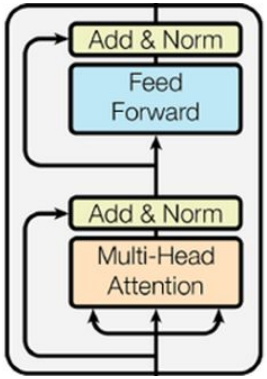




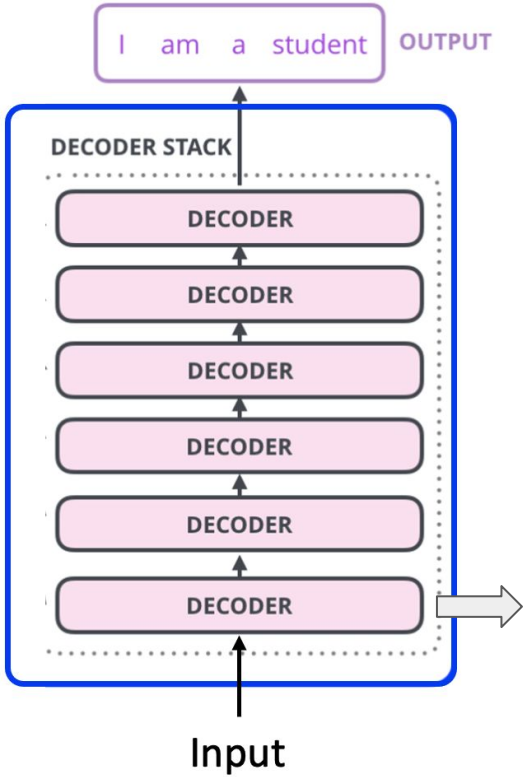
Vaswani, Ashish etal. Attention Is All You Need. NeurIPS 2017.

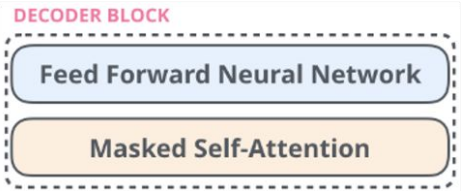
Transformer block

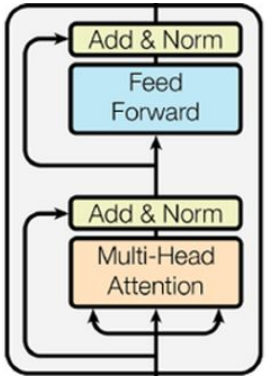




Decoder-based transformers



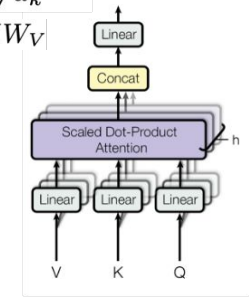




$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

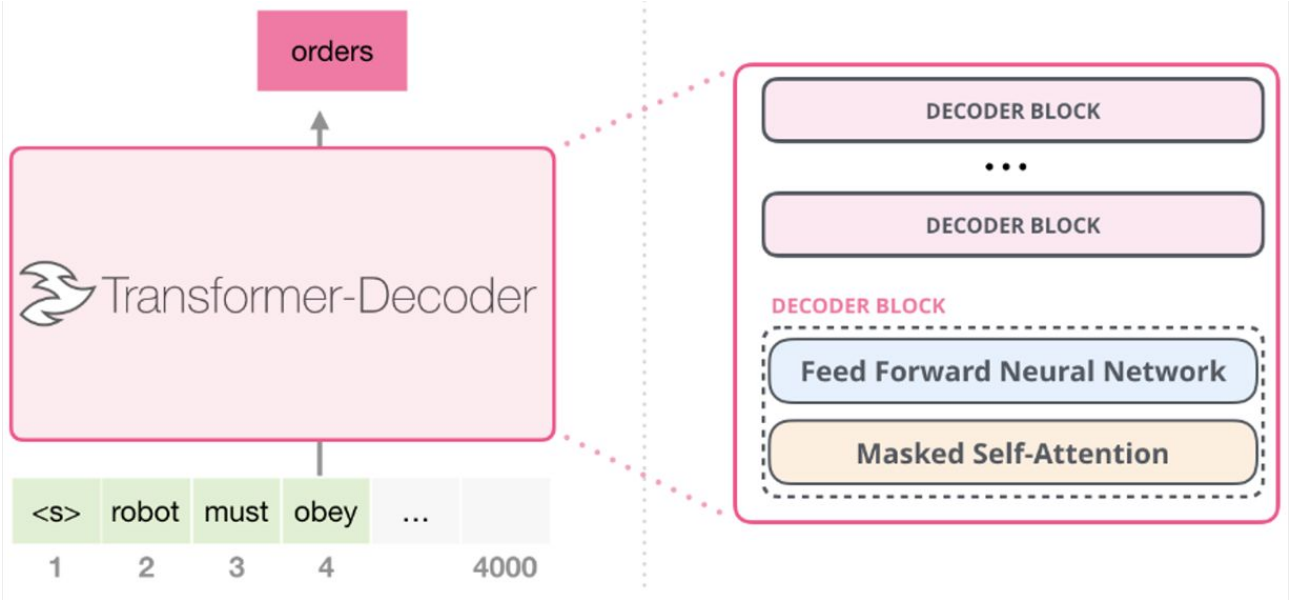
$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$



From Internet

GPT Architecture

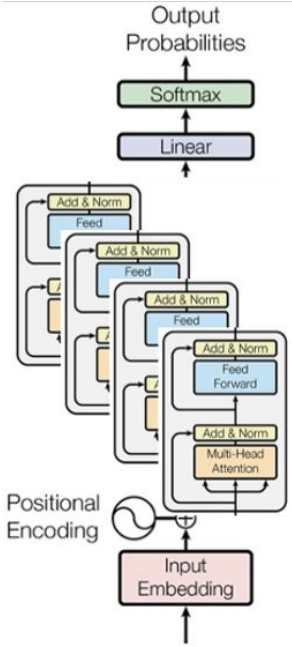
A stack of decoders (decoder blocks)



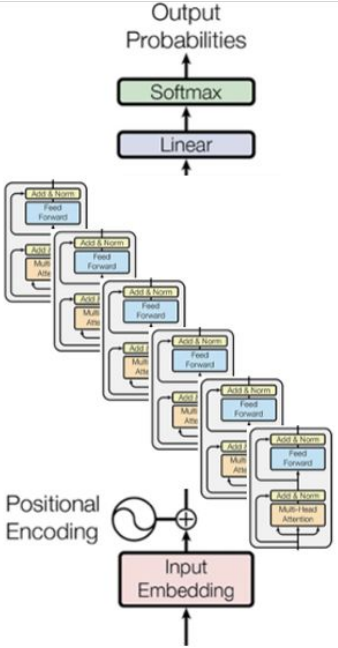
GPT models from OpenAI

GPT-2/-3/-4 have mostly just been larger versions. With the key differences coming from training data and training processes.

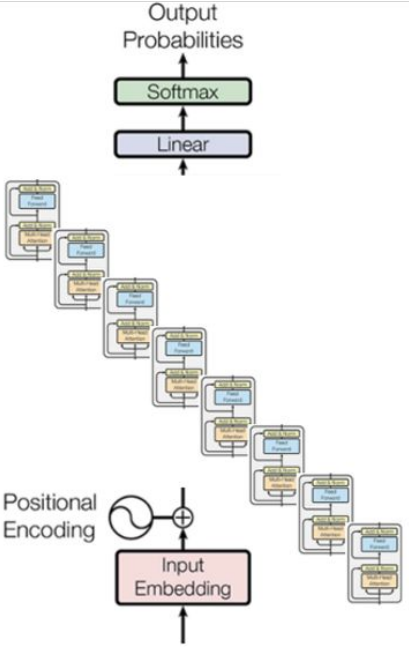
- GPT (2018)
- GPT-2 (2019)
- GPT-3 (2020)



GPT/GPT-1
12x
BooksCorpus

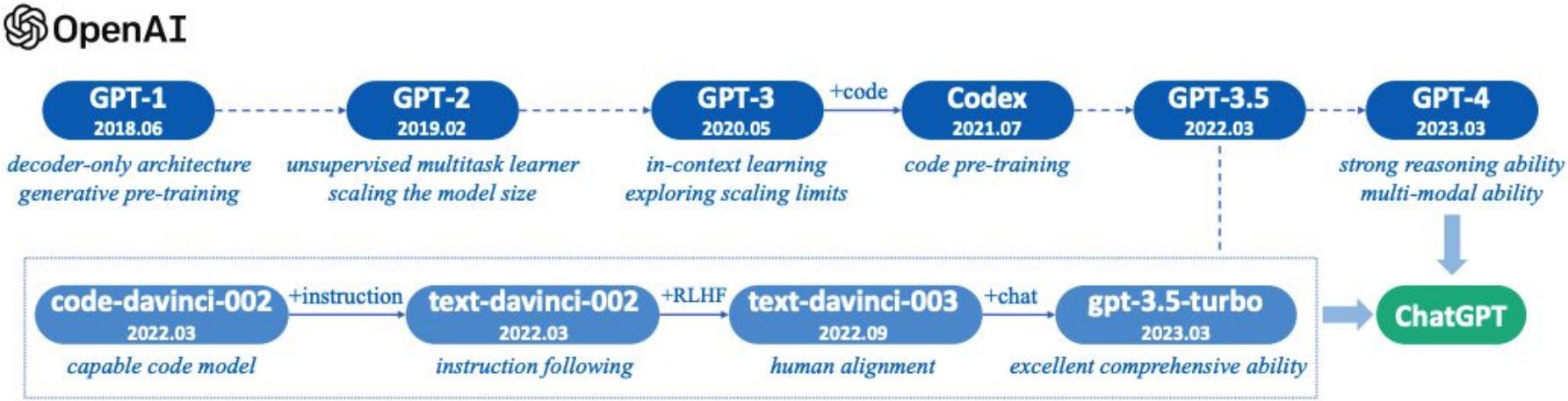


GPT-2
48x
WebText



GPT-3
96x
WebText 2

The rise of generative AI



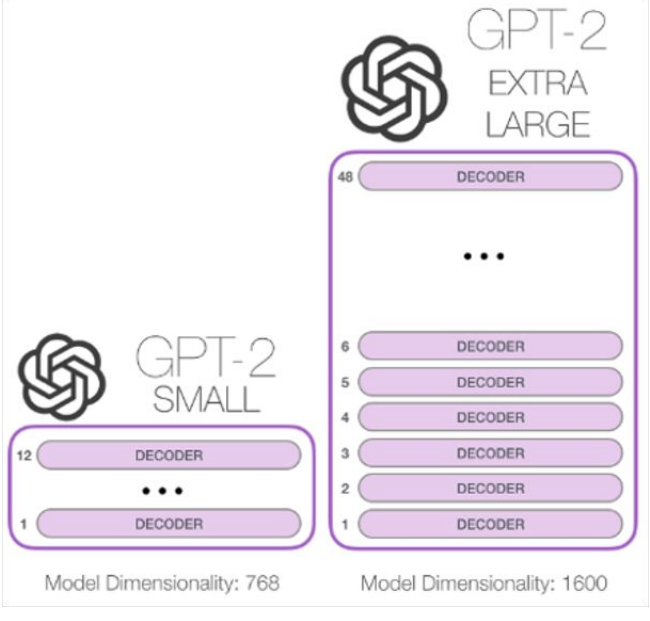
Zhao, Wayne Xin, et al. A survey of large language models. 2023.

GPT models have many parameters

Model size refers the number of parameters in the model.

The larger the model, the better it will be in understanding the nuances of human language.

Parameter	GPT-2 Small	GPT-2 Extra Large	Difference Extra Large / Small
Layers	12	48	3x
Model Dimensionality	768	1600	2x
Attention Heads per Layer	12	25	2.5x
Total	**117 Million**	**1.5 Billion**	**12.8x**



Prompt Engineering

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OxML 2025



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What is Prompt Engineering?

Prompt Engineering is the process of designing and optimising text inputs (prompts) to deliver consistent and quality responses (completions) for a given application objective and model.

Prompt Engineering

Prompt engineering is currently more art than science. The best way to improve our intuition for it is to practice more and adopt a trial-and-error approach that combines application domain expertise with recommended techniques and model-specific optimizations.

To execute the exercises you will need:

- An OpenAI API key - the service endpoint for a deployed LLM.
- A Python Runtime - in which the Notebook can be executed.

OPENAI_API_KEY=st-xxxxxxx

Prompt Engineering

We can think of this as a 2-step process:

- designing the initial prompt for a given model and objective
- refining the prompt iteratively to improve the quality of the response

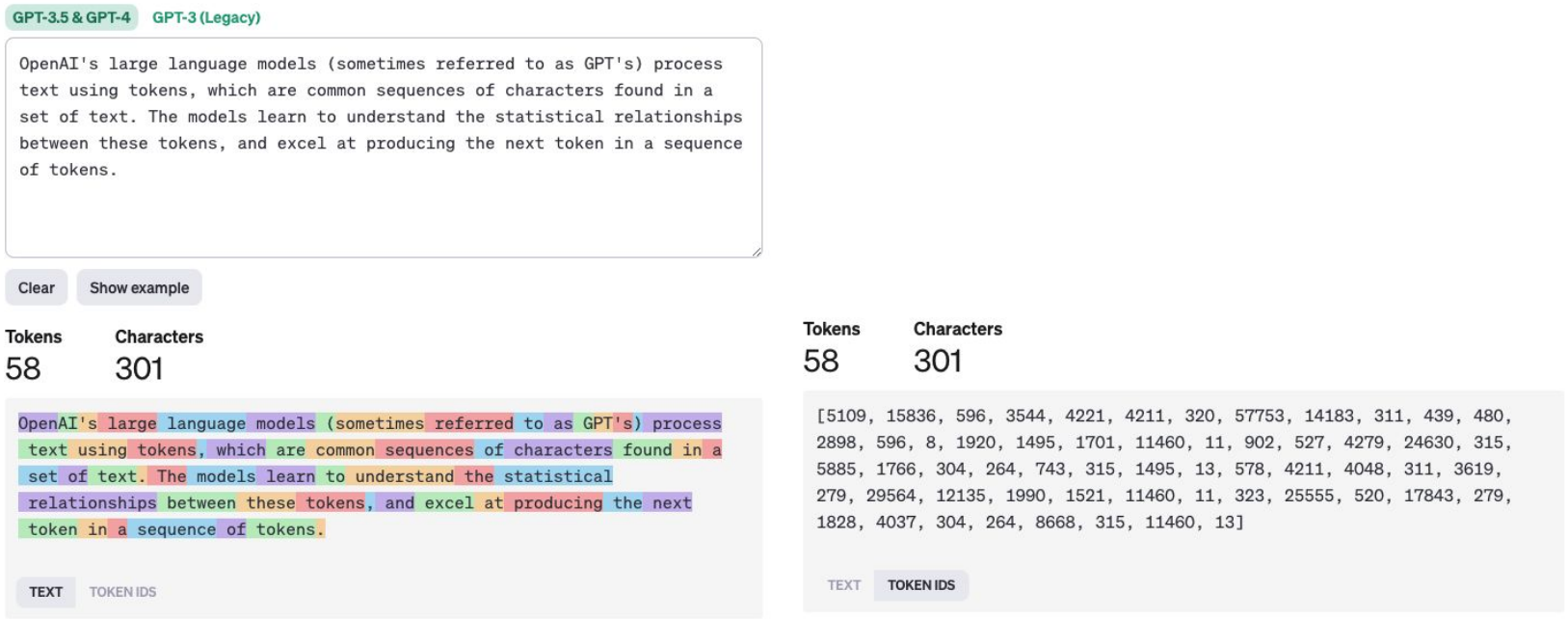
Trial-and-error process

We first need to understand three concepts:

- Tokenization = how the model "sees" the prompt
- Base LLMs = how the foundation model "processes" a prompt
- Instruction-Tuned LLMs = how the model can now see "tasks"

Tokenization

https://platform.openai.com/tokenizer



Foundation Models - LLMs

Once a prompt is tokenized, the primary function of the "Base LLM" (or Foundation model) is to predict the token in that sequence.

Since LLMs are trained on massive text datasets, they have a good sense of the statistical relationships between tokens and can make that prediction with some confidence. Note that they don't understand the meaning of the words in the prompt or token; they just see a pattern they can "complete" with their next prediction.

They can continue predicting the sequence till terminated by user intervention or some pre-established condition.

We can chat. But what if the user wanted to see something specific that met some criteria or task objective?

Instruction Tuned LLMs

An Instruction Tuned LLM starts with the foundation model and fine-tunes it with examples or input/output pairs (e.g., multi-turn "messages") that can contain clear instructions - and the response from the AI attempt to follow that instruction.

This uses techniques like Reinforcement Learning with Human Feedback (RLHF) that can train the model to follow instructions and learn from feedback so that it produces responses that are better-suited to practical applications and more-relevant to user objectives.

Instruction examples

Prompt:

```
Classify the text into neutral, negative or positive.
Text: I like walking.
Sentiment:
```

Output:

```
Positive
```

Instruction examples

Prompt:

Summarize the text delimited by triple backticks into a single sentence.
```You should express what you want a model to do by providing instructions that are as clear and specific as you can possibly make them. This will guide the model towards the desired output, and reduce the chances of receiving irrelevant or incorrect responses. Don't confuse writing a clear prompt with writing a short prompt. In many cases, longer prompts provide more clarity and context for the model, which can lead to more detailed and relevant outputs.
```

Output:

Clear and specific instructions are crucial when guiding a model's behavior, emphasizing the importance of expressing desired outcomes to reduce the likelihood of irrelevant or incorrect responses; the length of the prompt should not be sacrificed for clarity and context, as longer prompts often lead to more detailed and relevant model outputs.

Advanced Technologies

Zero-shot Prompting

Few-shot Prompting

Chain-of-Thought Prompting

Self-Consistency

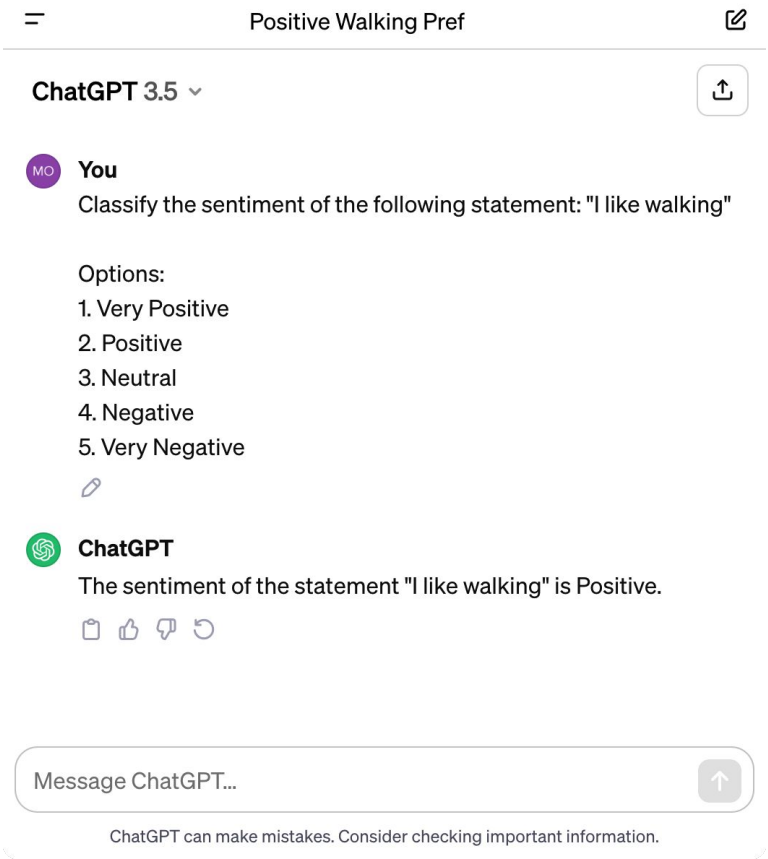
Zero-Shot Prompting

Prompt:

```
Classify the sentiment of the following statement: "{text}"

Options:
1. Very Positive
2. Positive
3. Neutral
4. Negative
5. Very Negative
```

*Replace {text} with the actual text you want to classify for sentiment.



Few-Shot Prompting

1-shot

3-shot

Classify the sentiment of the following sentences as 'Positive', 'Negative', or 'Neutral'.

Sentence: "I love sunny days."
Sentiment: Positive

Sentence: "Their performance exceeded all my expectations!"
Sentiment:

Classify the sentiment of the following sentences as 'Positive', 'Negative', or 'Neutral'.

Sentence: "I love sunny days."
Sentiment: Positive

Sentence: "This soup tastes terrible."
Sentiment: Negative

Sentence: "He is always so kind and helpful."
Sentiment: Positive

Sentence: "Their performance exceeded all my expectations!"
Sentiment:

Few-Shot Prompting

5-shot

Classify the sentiment of the following sentences as 'Positive', 'Negative', or 'Neutral'.

Sentence: "I love sunny days."
Sentiment: Positive

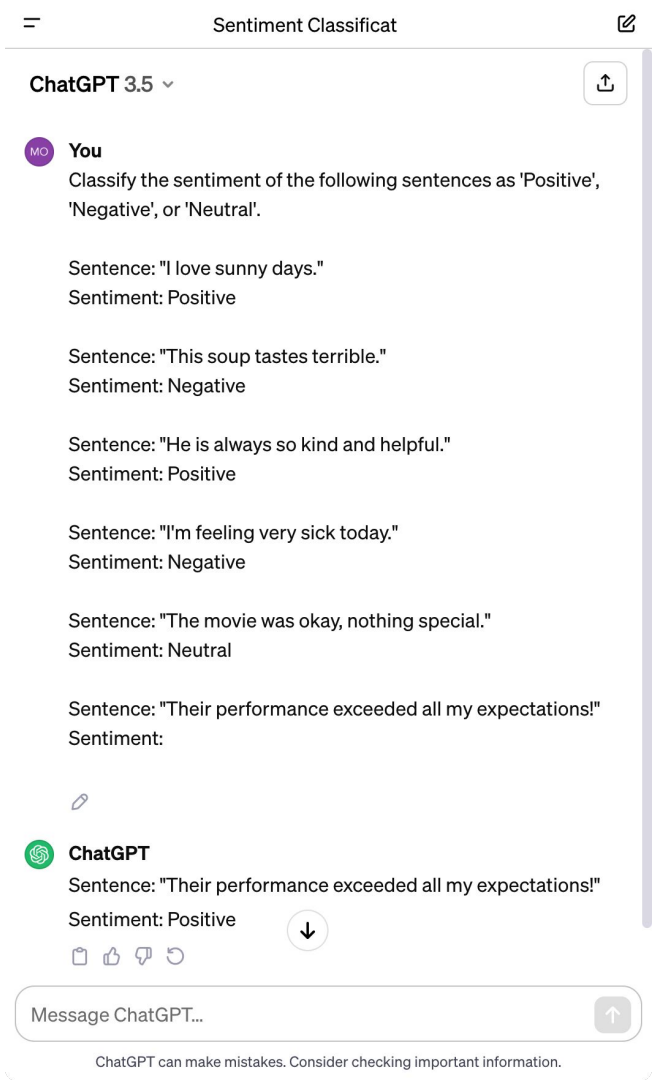
Sentence: "This soup tastes terrible."
Sentiment: Negative

Sentence: "He is always so kind and helpful."
Sentiment: Positive

Sentence: "I'm feeling very sick today."
Sentiment: Negative

Sentence: "The movie was okay, nothing special."
Sentiment: Neutral

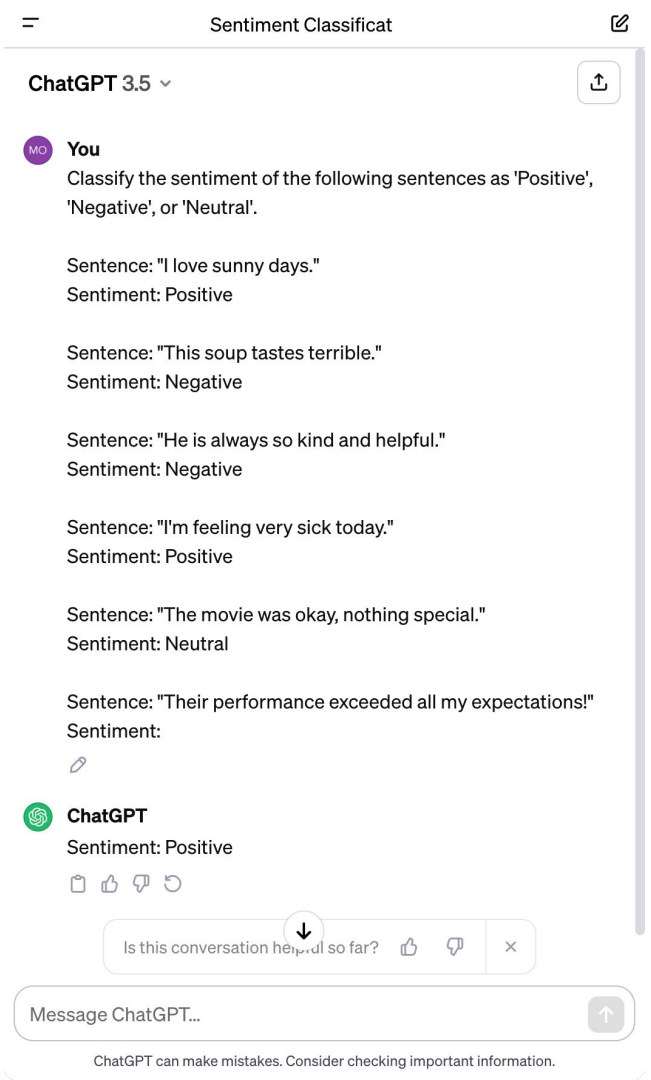
Sentence: "Their performance exceeded all my expectations!"
Sentiment:



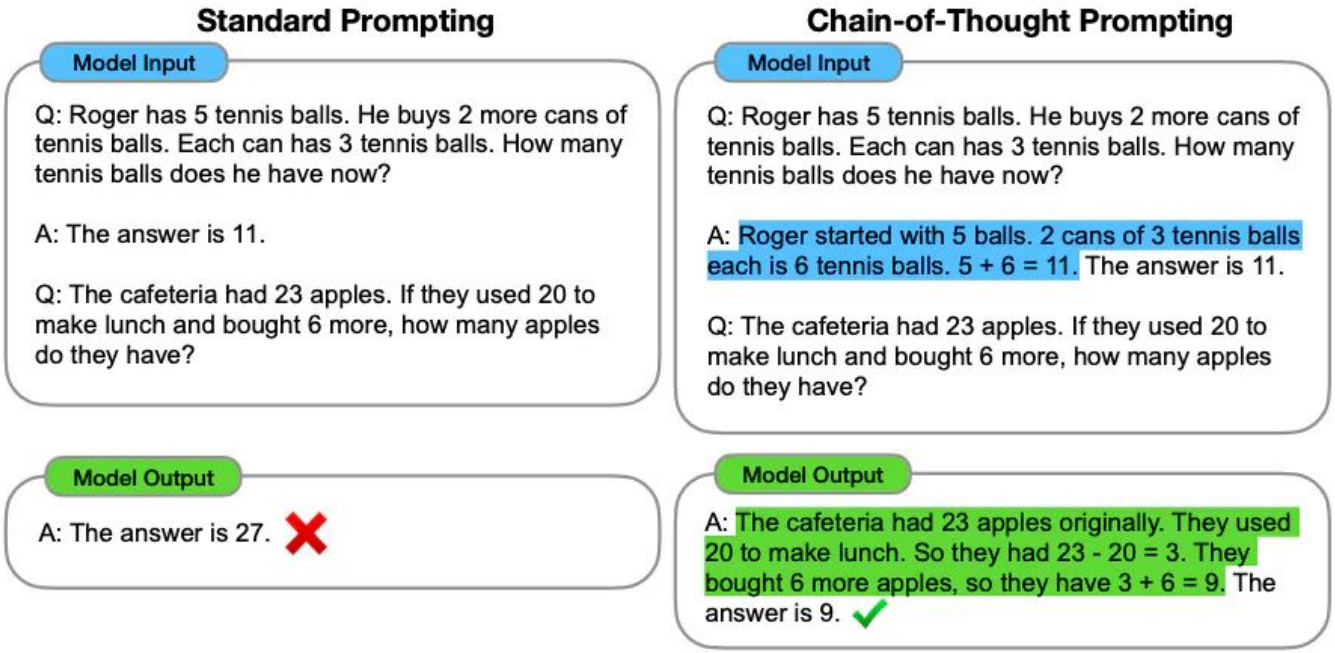
Some Tips for Few-Shot Prompting

- The label space and the distribution of the input text specified by the demonstrations are both important (regardless of whether the labels are correct for individual inputs).
- The format you use also plays a key role in performance, even if you just use random labels, this is much better than no labels at all.
- Additional results show that selecting random labels from a true distribution of labels (instead of a uniform distribution) also helps.

*random labels



Chain-of-Thought Prompting



Wei, Jason, et al. Chain-of-thought prompting elicits reasoning in large language models. 2022.

Chain-of-Thought Prompting

Prompt:

The odd numbers in this group add up to an even number: 4, 8, 9, 15, 12, 2, 1.
A: Adding all the odd numbers (9, 15, 1) gives 25. The answer is False.

The odd numbers in this group add up to an even number: 17, 10, 19, 4, 8, 12, 24.
A: Adding all the odd numbers (17, 19) gives 36. The answer is True.

The odd numbers in this group add up to an even number: 16, 11, 14, 4, 8, 13, 24.
A: Adding all the odd numbers (11, 13) gives 24. The answer is True.

The odd numbers in this group add up to an even number: 17, 9, 10, 12, 13, 4, 2.
A: Adding all the odd numbers (17, 9, 13) gives 39. The answer is False.

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1.
A:

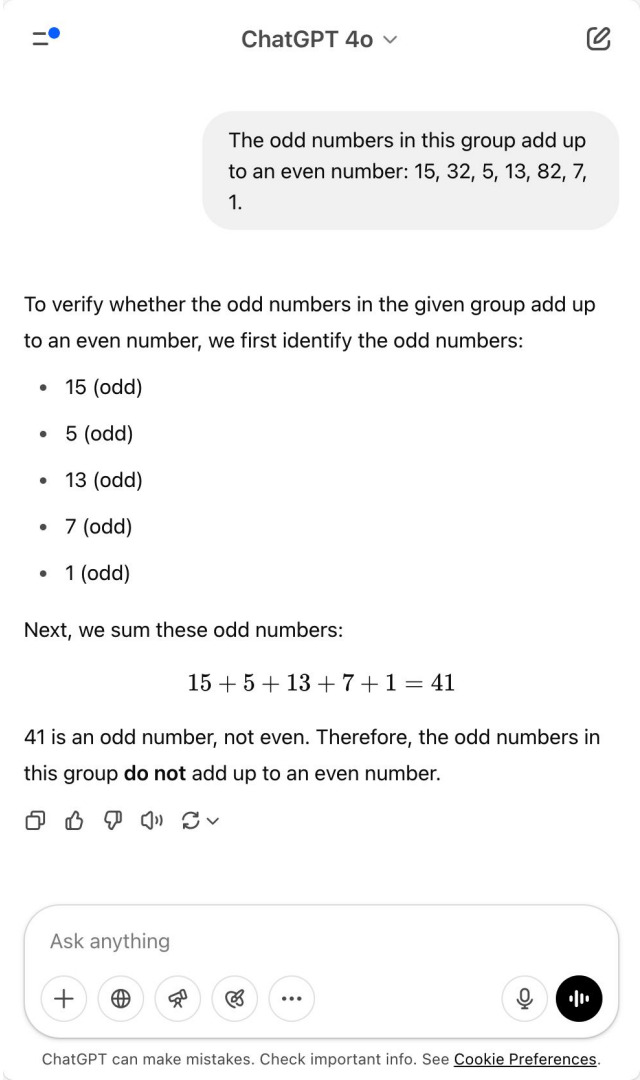
Output

Adding all the odd numbers (15, 5, 13, 7, 1) gives 41. The answer is False.

Chain-of-Thought Prompting

Prompt via reasoning models:

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1.



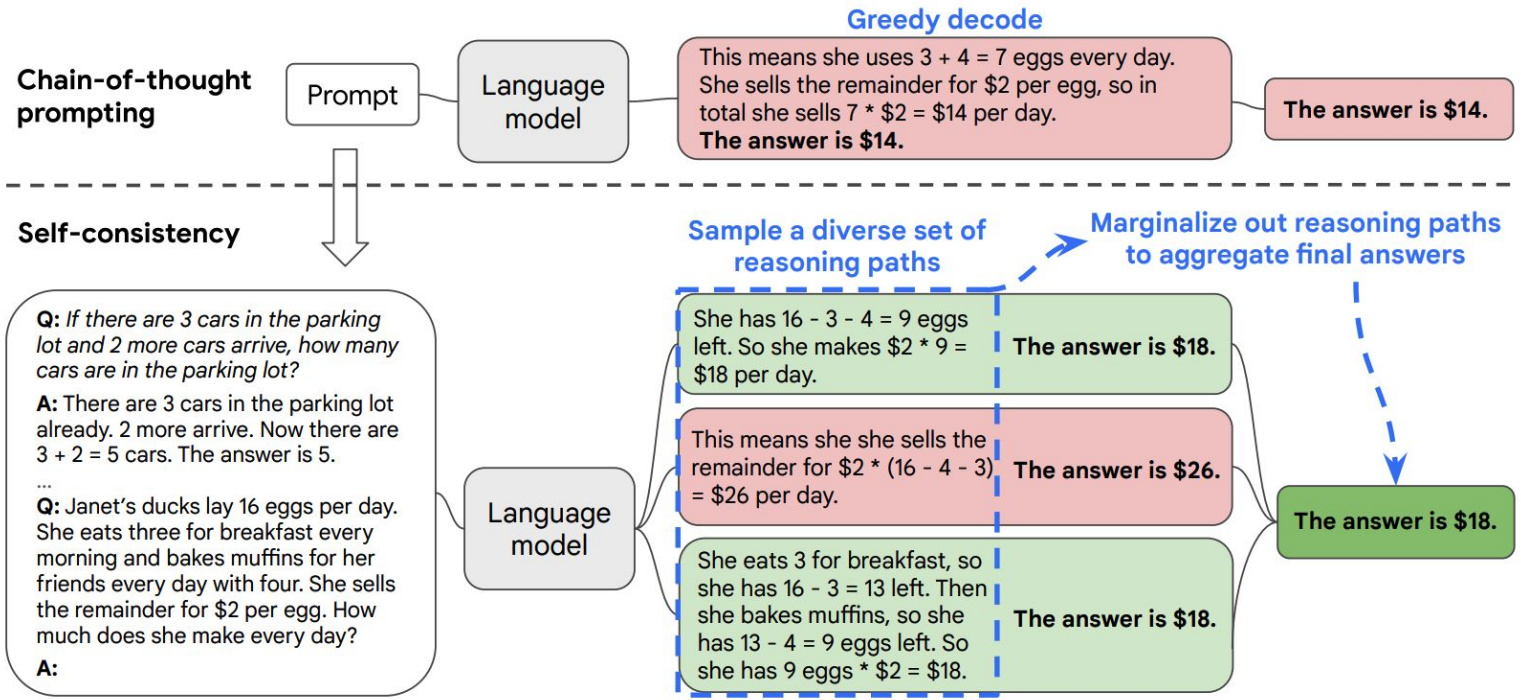
Self-Consistency

Self-consistency aims to replace the naive greedy decoding used in chain-of-thought prompting.

The idea is to sample multiple, diverse reasoning paths through few-shot CoT, and use the generations to select the most consistent answer.

This helps to boost the performance of CoT prompting on tasks involving arithmetic and commonsense reasoning.

Chain-of-thought vs. Self-Consistency



Wang, Xuezhi, et al. "Self-consistency improves chain of thought reasoning in language models." arXiv preprint arXiv:2203.11171 (2022).

Expanding Knowledge for LLMs

Meng Fang
OxML 2025



UNIVERSITY OF LIVERPOOL

Retrieval Augmented Generation (RAG)

RAG combines an information retrieval component with a text generator model. RAG can be fine-tuned and its internal knowledge can be modified in an efficient manner and without needing retraining of the entire model.

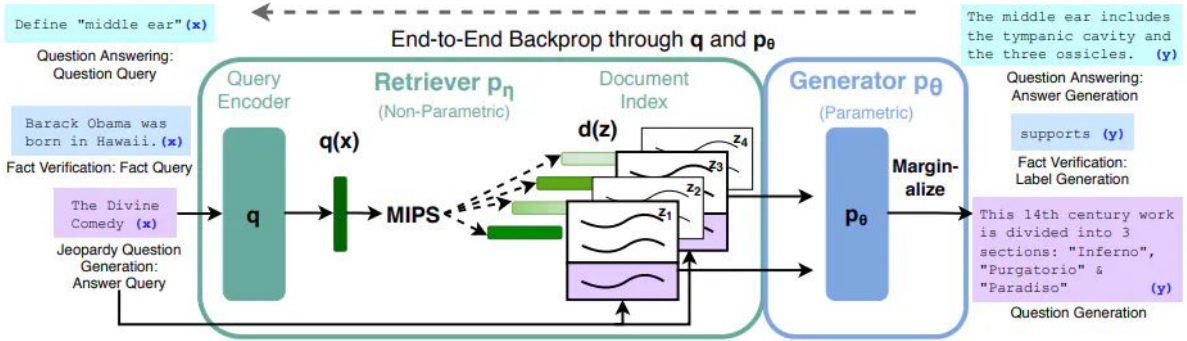


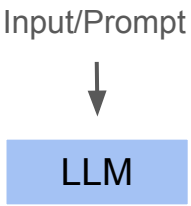
Figure 1: Overview of our approach. We combine a pre-trained retriever (*Query Encoder + Document Index*) with a pre-trained seq2seq model (*Generator*) and fine-tune end-to-end. For query x, we use Maximum Inner Product Search (MIPS) to find the top-K documents z_i. For final prediction y, we treat z as a latent variable and marginalize over seq2seq predictions given different documents.

Lewis, Patrick, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. Advances in Neural Information Processing Systems. 2020.

Retrieval-Augmented Generation

Generation: Large Language Models (LLMs), such as ChatGPT/GPT-3.5/4 and others

LLMs can generate texts in response to a user input, referred to as a prompt.



U

Who won the world cup 2018?

The 2018 FIFA World Cup was won by the French national football (soccer) team. France defeated Croatia 4-2 in the final match to win their second FIFA World Cup. The tournament took place in Russia from June 14 to July 15, 2018.

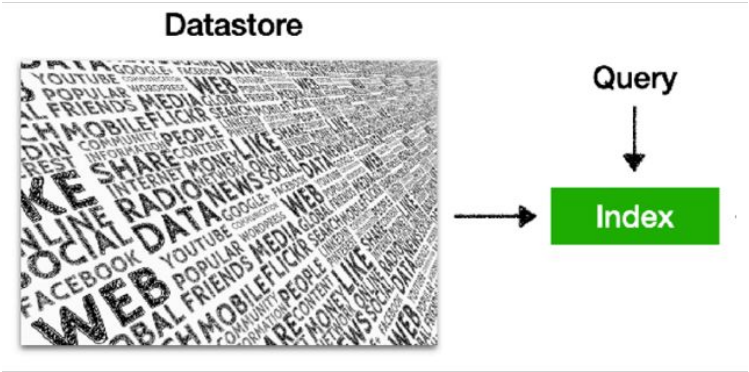
U

Who won the world cup 2022?

I'm sorry, but as of my last knowledge update in September 2021, I do not have information on the winner of the 2022 FIFA World Cup, as it had not yet taken place. The 2022 FIFA World Cup was scheduled to be held in Qatar from November 21 to December 18, 2022. To find out the winner of the tournament, I recommend checking the latest news or conducting an online search for up-to-date information.

Retrieval

Retrieval from a datastore or a collection of documents

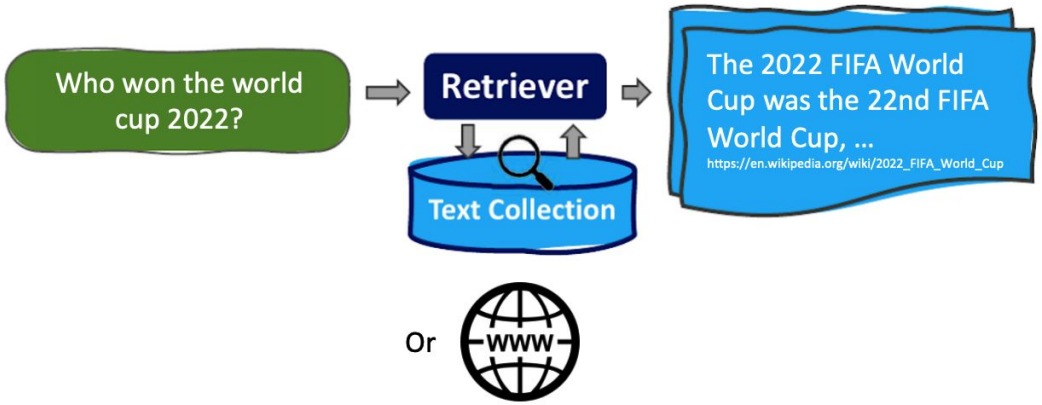


Internet,
or a collection of documents

- Representative tasks: open-domain QA, fact checking, entity linking, ..

Retrieval

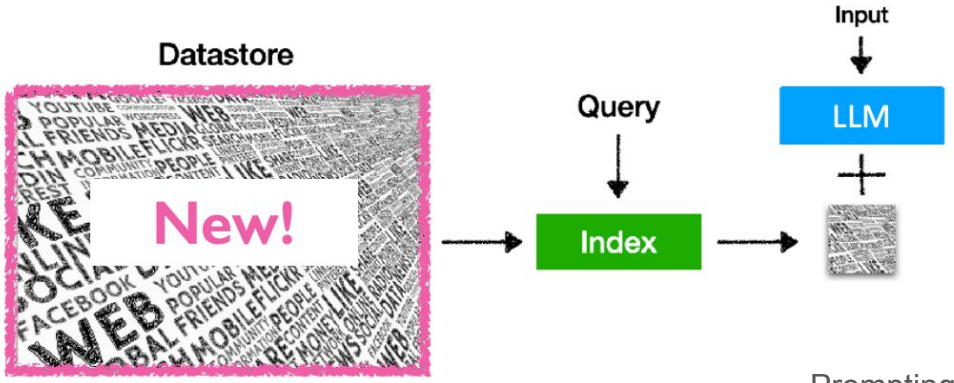
Retrieval from a datastore or a collection of documents



- Representative tasks: open-domain QA, fact checking, entity linking, ..

Retrieval-Augmented Generation

Retrieval + LLM



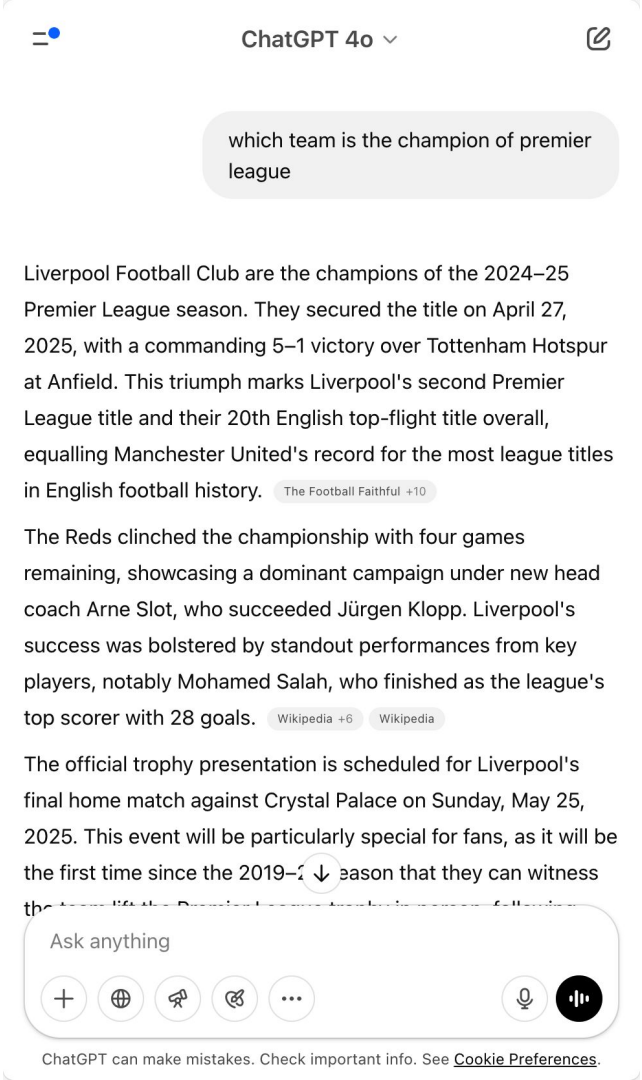
Prompting:
- Instruction
- Retrieved content
- Question

Asai, Akari, et al. Retrieval-based Language Models and Applications. ACL 2023.

Retrieval-Augmented Generation

Retrieval + LLM

GPT-4o can support searching websites.



Why retrieval → LLMs?

LLMs’ knowledge is easily outdated and hard to update

LLMs are shown to easily leak private training data and use local data

LLMs are *large* and expensive to train and run

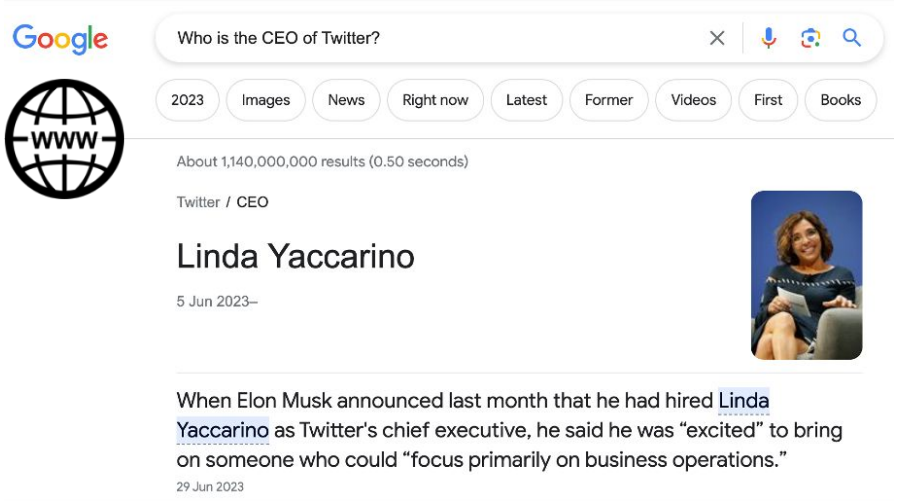
…

LLMs' knowledge is easily outdated and hard to update

Who is the CEO of Twitter?

ChatGPT

As of my **knowledge cutoff in September 2021**, the CEO of Twitter is **Jack Dorsey**....



The datastore can be easily updated and expanded - even without retraining!

LLMs are shown to easily leak private training data and use local data

GA Can you write the first page of "True History of the Kelly Gang"

Certainly! Here is the first page of Peter Carey's "True History of the Kelly Gang":

I, NED KELLY, of the Fenian Brotherhood, am the adopted son of a man called Red Kelly and I am the state's most wanted bushranger. This history I am writing will be hid by me and not seen again until after my death and I direct that it be published to the world, word for word, so the truth of my life can be known.

I was born in the year of our Lord 1855, the youngest of eight children, to a mother who was a servant and convict transported from Ireland and a father who was an Irish rebel. I was raised in poverty and ignorance in the colony of Victoria. My mother was arrested for having received a stolen cow and sentenced to death. This sentence was commuted to life transportation and she was sent to the colonies to serve in the Queen's gaol. My father was a wild colonial and in his blood there was a fighting fire. He was shot dead by the police when I was twelve years old and I can still see him falling.

Danger! Copyrighed text.

Private datastore / data access control



LLMs are *large* and expensive to train and run

Can we train LLMs every week?

Can we possibly reduce the training and inference costs, and scale down the size of LLMs?

- Use datastore and retrieval

Fine-tuning

To carry out a task that isn’t easy to define in a prompt.

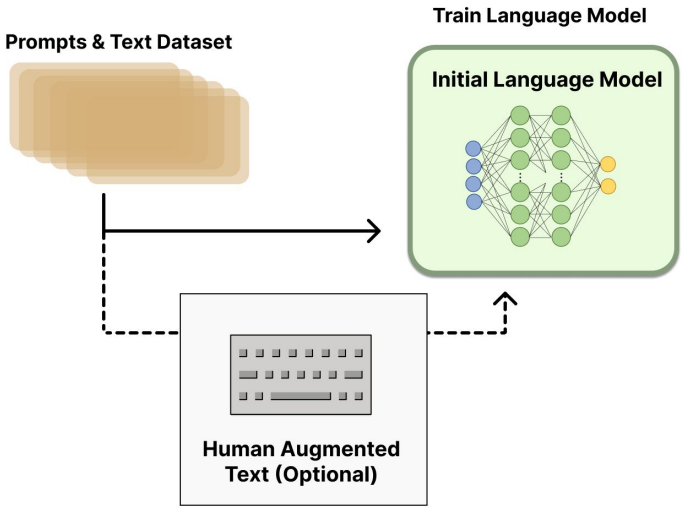
To help LLM gain specific knowledge.

To get a smaller model to perform a task.

Pretraining an LLM

Pretrain general-purpose LLMs by learning from internet text.

For building a specific application: option of last resort; could help if have a highly specialized domain



From Internet

Choosing a model

Model size

1B parameters: GPT-Neo (1.3B), Pythia-1B, OPT-1.3B

10B parameters: LLama-13B, Mistra-13B, …

100B parameter: GPT-3.5, GPT-4

Closed or open source?

Closed-source models

- Easy to use in applications
- More large/powerful models
- Relatively inexpensive
- Some risk of vendor lock-in

Open-source models

- Full control over model
- Can run on your own device
- Full control over data privacy/access

How do chat systems learn to follow instructions?

Fine tuning

Help me brainstorm some fun museums in London.

Here are some suggestions …

Input (instructions) -> Output

Help me brainstorm some fun museums in London. -> Here

Help me brainstorm some fun museums in London. Here -> are some suggestions

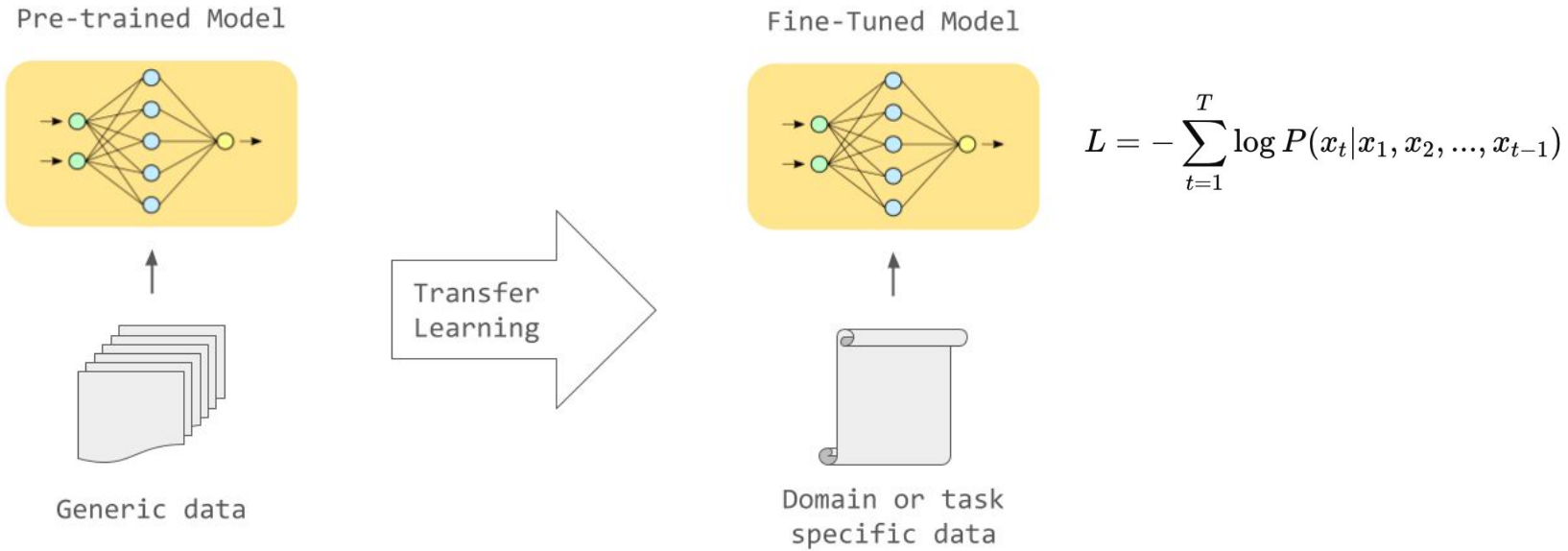
Help me brainstorm some fun museums in London. Here are -> some suggestions

Help me brainstorm some fun museums in London. Here are some -> suggestions

… -> …

How do chat systems learn to follow instructions?

Cross-Entropy Loss for Instruction Tuning



$$L = -\sum_{t=1}^{T} \log P(x_t | x_1, x_2, ..., x_{t-1})$$

Reinforcement learning from human feedback

Step 1: Train a reward (quality) model

Step 2: Have LLM generate a lot of answers. Further train it to generate more responses that get high scores

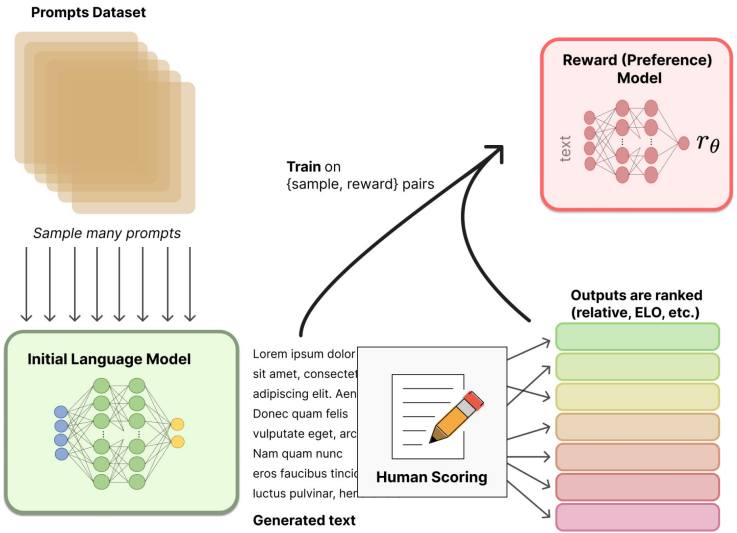
Input -> Output

(Response) | (Score/Reward)

I’m happy to help. Here are some steps … | 5

Just try your best! | 3

It is hopeless! | 1

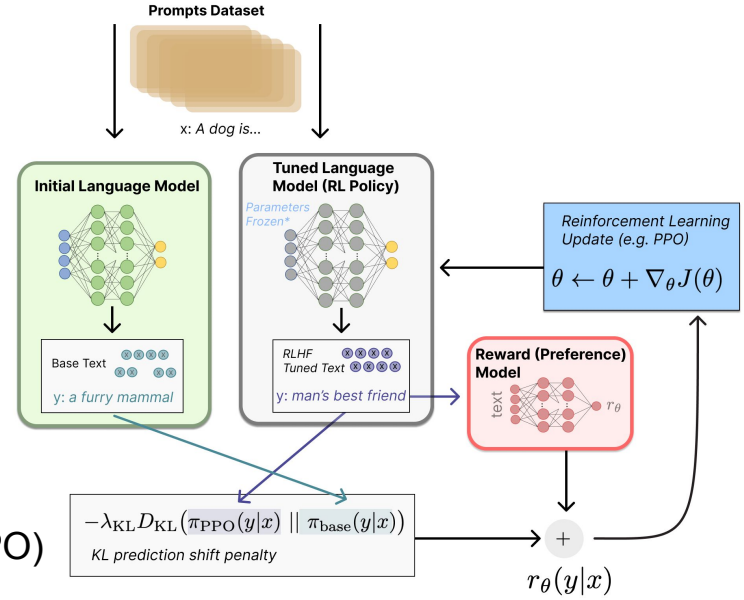


From Internet

Reinforcement learning from human feedback

Step 1: Train a reward (quality) model

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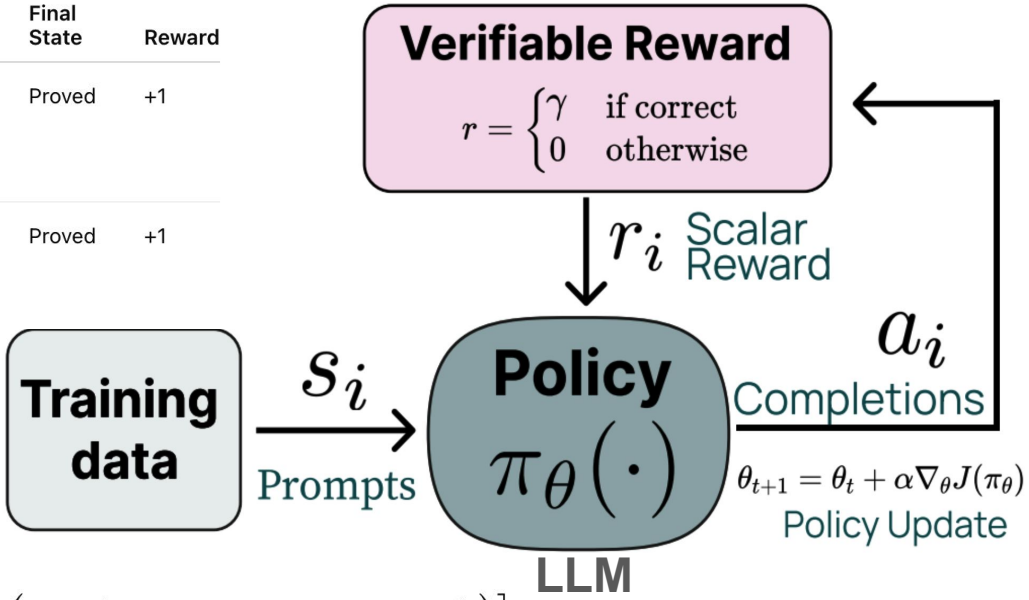


Proximal Policy Optimization (PPO)

From Internet

Reinforcement learning from verifiable rewards

Problem	Initial State	Valid Actions	Expected Proof	Final State	Reward
Problem 1	$A + B = B + A$	Apply commutative law	$B + A = A + B$	Proved	+1
Problem 2	$A \times (B + C) = (A \times B) + (A \times C)$	Apply distributive law	$(A \times B) + (A \times C)$	Proved	+1



$$L_{\text{policy}}(\theta) = \mathbb{E}\left[\min\left(r_t(\theta)\hat{A}_t, \text{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t\right)\right]$$

where:

- $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio between the new and old policies.

LLM Agents

Meng Fang
OxML 2025



UNIVERSITY OF LIVERPOOL

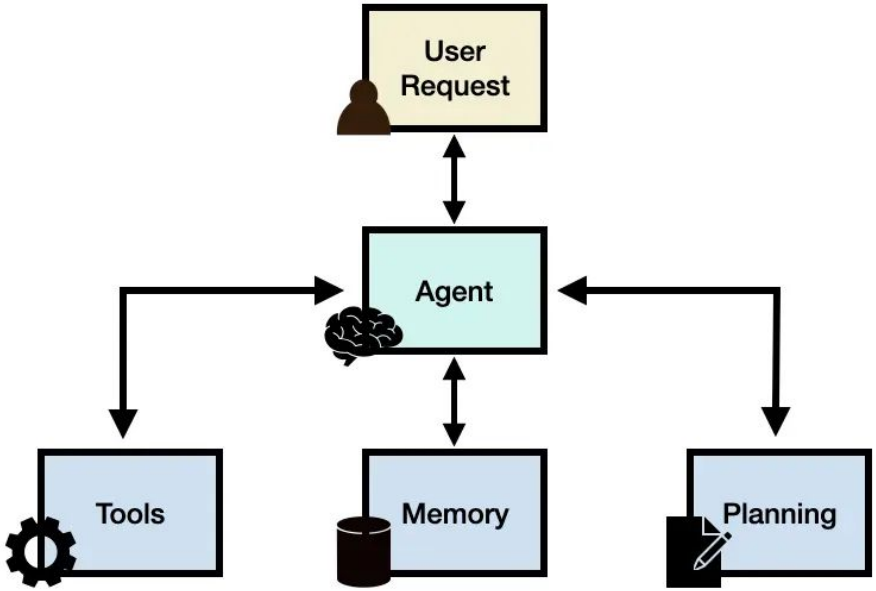
Agent System

User Request - a user question or request

Agent/Brain - the agent core acting as coordinator

Planning - assists the agent in planning future actions

Memory - manages the agent's past behaviors



Agent Applications

Reporting: Analyzing vast amounts of data and generating comprehensive reports.

Research: Conducting in-depth investigations across various domains, such as legal, finance, and health.

Booking: Assisting with travel arrangements and event planning.

Financial analysis: Analyzing market trends, assess financial data, and generate reports with unprecedented speed and accuracy.

Memory Systems: Retaining and Utilizing Information

- Short-term (Working) Memory
  - Functions as a buffer for immediate context
  - Enables in-context learning
  - Sufficient for most task completions
  - Helps maintain continuity during task iteration
- Long-term Memory
  - Implemented through external vector stores
  - Enables fast retrieval of historical information
  - Valuable for future task completion
  - Less commonly implemented but potentially crucial for future developments
- Memory systems allow agents to store and retrieve information gathered from external tools, enabling iterative improvement and building upon previous knowledge.

Agent System

Planning

- Decomposition

Reasoning

Tool use

- The agent learns to call external APIs for extra information that is missing from the model weights (often hard to change after pre-training)

Example:

Use LLM to choose and carry out complex sequences of actions

Help me research BurgerKing to competitors:

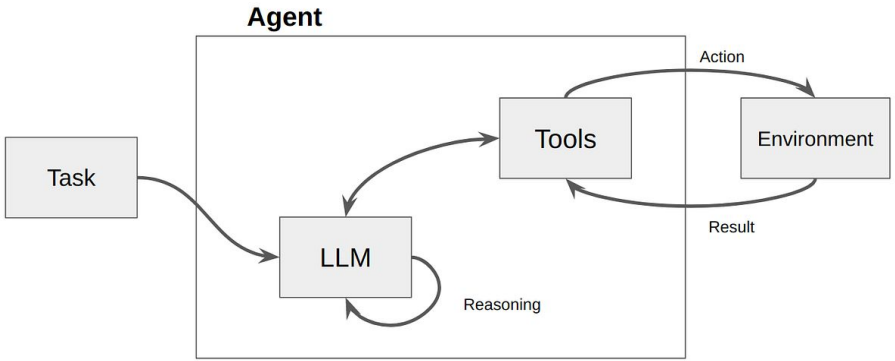
1. Search top competitors
2. Visit website of each competitor
3. For each competitor, write summary based on homepage content



SEARCH(BurgerKing competitors)
VISIT(https://www.kfc.com/)
VISIT(https://www.mcdonalds.com/)
…
Summarise the following text:
At BurgerKing, we pride ourselves on customizing their meals …

Agent - Tool Use

The model is allowed to access external tools—by predicting special tokens or commands. A tool can take various forms: it can be
a) the model itself or another neural network;
b) a retrieval component such as a search engine;
c) a symbolic computation or code module; or
d) a module for controlling a physical robot or virtual agent as in the previous newsletter:



Tool

Tools for reasoning: decide which APIs to call, when to call them, what arguments to pass, and how to best incorporate the results into future token prediction

[Prompt] $x_{1:i-1}$ -> <API>

The New England Journal of Medicine is a registered trademark of **[QA("Who is the publisher of The New England Journal of Medicine?") → Massachusetts Medical Society]** the MMS.

Out of 1400 participants, 400 (or **[Calculator(400 / 1400) → 0.29]** 29%) passed the test.

The name derives from "la tortuga", the Spanish word for **[MT("tortuga") → turtle]** turtle.

The Brown Act is California's law **[WikiSearch("Brown Act") → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.]** that requires legislative bodies, like city councils, to hold their meetings open to the public.

Schick, Timo, et al. "Toolformer: Language models can teach themselves to use tools." arXiv preprint arXiv:2302.04761 (2023).

Agent - Planning



Browse Internet



Empty Dishwasher



Turn off TV



Organize Closet



Wash Face



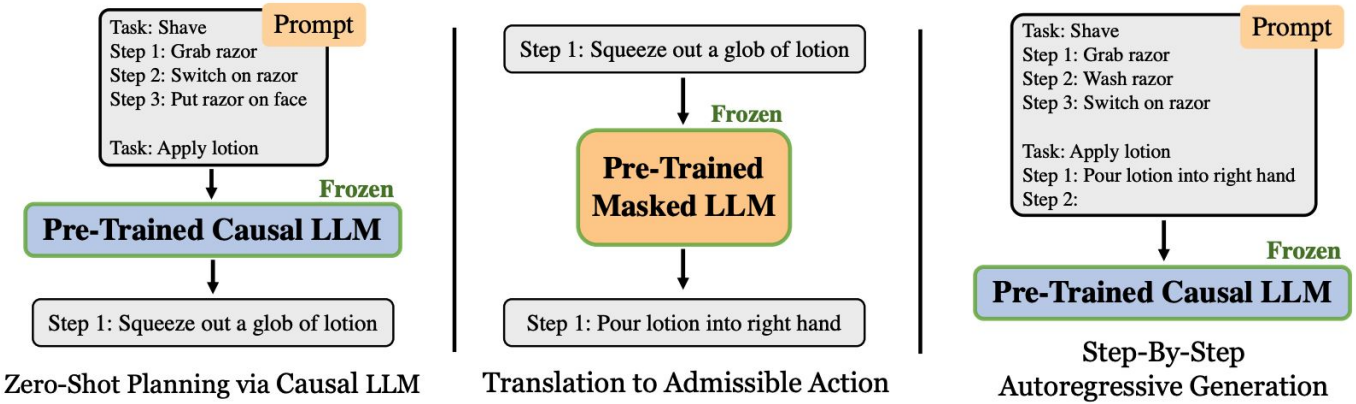
Take off Shoes

Huang, Wenlong, et al. Language Models as Zero-Shot Planners: Extracting Actionable Knowledge for Embodied Agents (ICML 2022).

Planning

Decompose high-level tasks into sensible mid-level action plans

Each step is admissible action

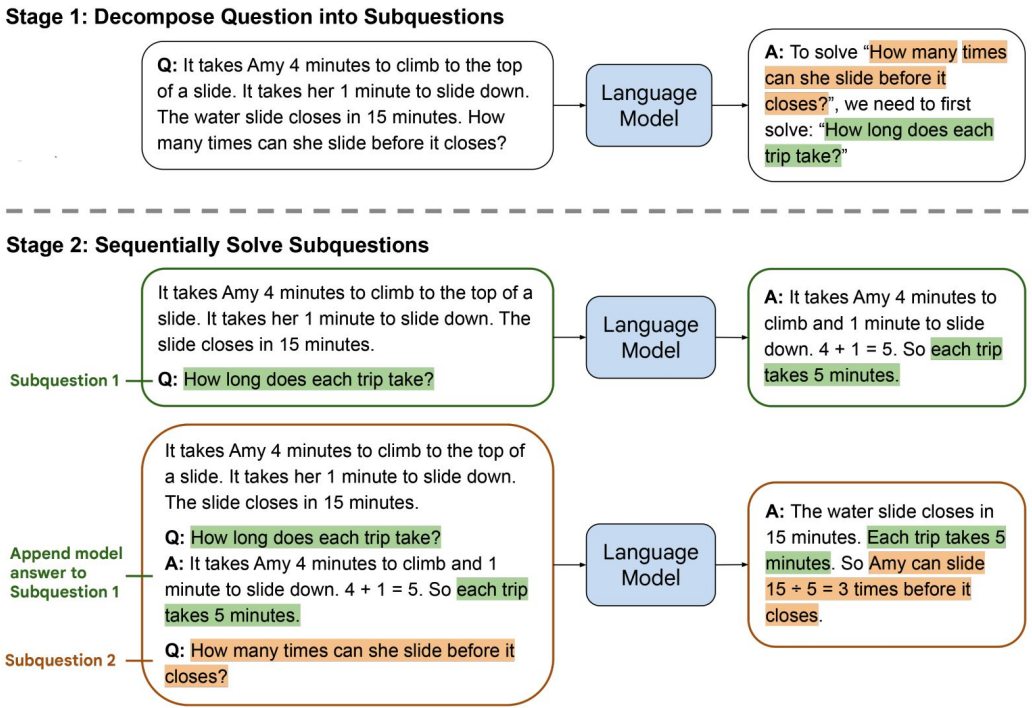


Large Language Models (LLMs) such as GPT-3 and Codex can plan actions for embodied agents, even without any additional training.

Huang, Wenlong, et al. Language Models as Zero-Shot Planners: Extracting Actionable Knowledge for Embodied Agents (ICML 2022).

Planning

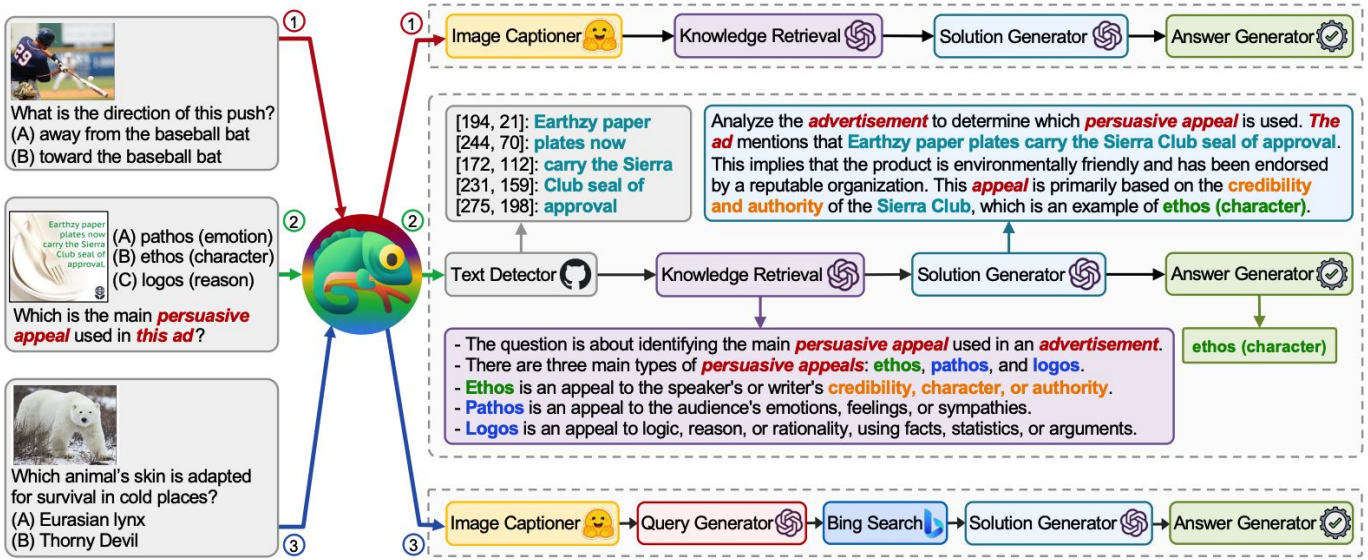
Decomposition



Zhou, Denny, et al. LEAST-TO-MOST PROMPTING ENABLES COMPLEX REASONING IN LARGE LANGUAGE MODELS (ICLR 2023).

Planning and Reasoning

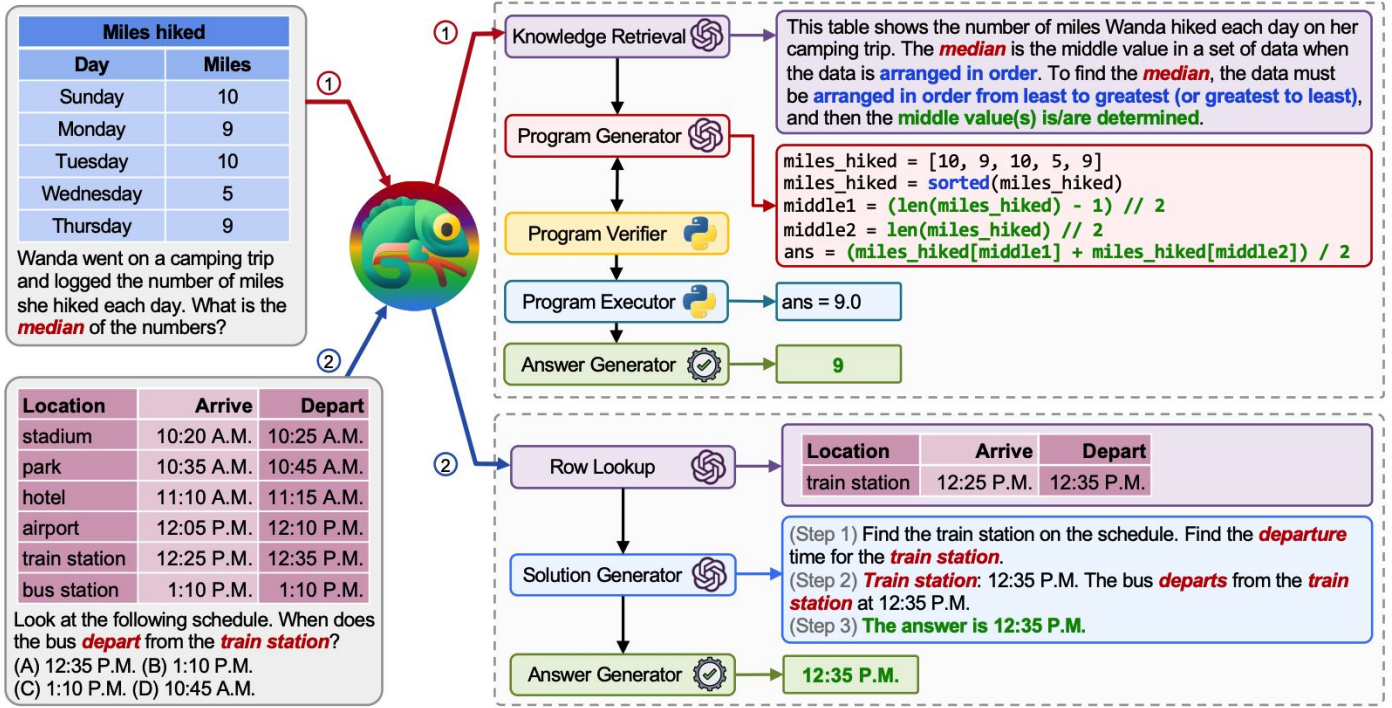
Synthesize programs to compose various tools and executing them sequentially to get final answers



Lu, Pan et al. Chameleon: Plug-and-Play Compositional Reasoning with Large Language Models (NeurIPS 2023).

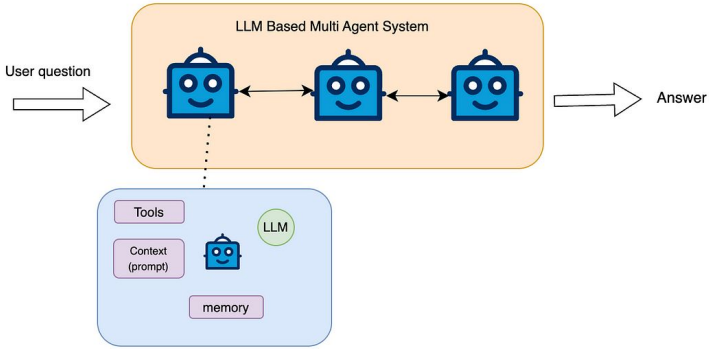
Planning, Reasoning, Tool use

Examples



Multi-agent systems via LLMs





Frameworks that are used to build LLM agents:

- LangChain: a framework for developing applications and agents powered by language models.
- AutoGPT: provides tools to build AI agents.
- AutoGen: a framework that enables the development of LLM applications using multiple agents that can converse with each other to solve tasks.
- LlamaIndex - a framework for connecting custom data sources to large language models.
- GPT Researcher: an autonomous agent designed for comprehensive online research on a variety of tasks.
- AgentVerse: designed to facilitate the deployment of multiple LLM-based agents in various applications.
- crewAI: AI agent framework reimagined for engineers, offering powerful capabilities with simplicity to build agents and automations.
- Phidata: a toolkit for building AI Assistants using function calling.

Summary

Introduction to Generative AI and LLMs

Prompt Engineering

Expanding Knowledge for LLMs

LLM Agents

Thank you.

Q & A

Course page: https://github.com/mengfn/generative-ai-nlp